

Space Weather®

RESEARCH ARTICLE

10.1029/2025SW004900

Key Points:

- A convolutional neural network with a multi-head attention mechanism forecasts geomagnetically induced currents (GICs)
- Forecasting geoelectric fields before GIC calculation improves prediction accuracy compared with direct GIC forecasting
- Deep learning-based GIC forecasts outperform GIC estimates based on one-dimensional conductivity and magnetotelluric impedance data

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

W. Kong and Z. Zhou,
wkong90@foxmail.com;
zhouzk2017@outlook.com

Citation:

Zhang, X., Pi, G., Kong, W., Zhou, Z., Li, S., Němeček, Z., et al. (2026). Forecasting geomagnetically induced currents from solar wind and geoelectric field data using deep learning: Application in Fujian, China. *Space Weather*, 24, e2025SW004900. <https://doi.org/10.1029/2025SW004900>

Received 16 DEC 2025

Accepted 14 SEP 2026

Author Contributions:

Conceptualization: Xuejian Zhang, Wenxin Kong, Nian Yu

Data curation: Gilbert Pi

Formal analysis: Zdeněk Němeček, Ye Fan

Funding acquisition: Wenxin Kong, Zikun Zhou, Nian Yu

Investigation: Gilbert Pi

Methodology: Xuejian Zhang

Project administration:

Zdeněk Němeček, Jana Šafránková

© 2026 The Author(s). Space Weather published by Wiley Periodicals LLC on behalf of American Geophysical Union. This is an open access article under the terms of the [Creative Commons Attribution-NoDerivs License](#), which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

Forecasting Geomagnetically Induced Currents From Solar Wind and Geoelectric Field Data Using Deep Learning: Application in Fujian, China

Xuejian Zhang¹ , Gilbert Pi² , Wenxin Kong^{3,4} , Zikun Zhou^{3,4} , Sheng Li² , Zdeněk Němeček² , Jana Šafránková² , Ye Fan⁵ , and Nian Yu^{1,4} 

¹School of Electrical Engineering, Chongqing University, Chongqing, China, ²Faculty of Mathematics and Physics, Charles University, Prague, Czech Republic, ³School of Resources and Safety Engineering, Chongqing University, Chongqing, China, ⁴State Key Laboratory of Coal Mine Disaster Dynamics and Control, Chongqing University, Chongqing, China, ⁵China Earthquake Network Center, Beijing, China

Abstract We developed a multi-module convolutional neural network combined with a multi-head attention mechanism (MM-CNN-MHA) to forecast geomagnetically induced currents (GICs) at the Huangmeishan (HMS) 220-kV substation in Fujian Province, China. The model was trained using solar wind parameters, geomagnetic disturbances, and geoelectric field observations from 2010 to 2021. Five GIC estimation or prediction methods were evaluated: (a) GICs calculated from geoelectric fields predicted by MM-CNN-MHA; (b) direct GICs prediction using MM-CNN-MHA; (c) GICs calculated from measured geoelectric fields; (d) GICs calculated using a one-dimensional (1D) conductivity model; and (e) GICs calculated using magnetotelluric (MT) impedance data. The results show that the GICs calculated from measured geoelectric fields achieved the highest accuracy, with a root-mean-square error (RMSE) of 0.24 A, and better reproduced the observed GICs peaks. The GICs calculated from MM-CNN-MHA-predicted geoelectric fields outperformed the directly predicted GICs, with RMSE values of 0.26 and 0.28 A, respectively. In contrast, the GICs calculated using the 1D conductivity model and MT impedance method generally overestimate GICs peaks, with RMSE values of 0.52 and 0.39 A, respectively. These findings further highlight the importance of local geoelectric field observations for GIC monitoring and forecasting and demonstrate that preserving the physical chain from solar wind, geomagnetic disturbances, and geoelectric fields to GICs improves the stability and reliability of GIC predictions.

Plain Language Summary Space weather events can cause electric currents to flow in power systems, potentially affecting the safe operation of transformers and transmission lines. These currents are known as geomagnetically induced currents (GICs). Understanding and predicting them is important for reducing potential risks to power grids. In this study, we use a machine learning method to forecast GICs at power substations in Fujian Province, China. The model uses solar wind data measured at the Lagrange-1 (L1) point, along with electric field data measured on the ground (which drives these currents), to predict how strong these currents may become. We test two different approaches: one that first predicts the electric field in the ground and then calculates the currents, and another that directly predicts the currents. We also examine which approach is more effective and highlight the challenges and limitations in predicting these currents.

1. Introduction

Solar activity interacting with the Earth's magnetosphere can trigger geomagnetic storms, which subsequently induce geoelectric fields at the Earth's surface and generate geomagnetically induced currents (GICs) in power grids (Beggan et al., 2021; Cordell et al., 2024; Liu et al., 2024). GICs typically enter the power grid through transformer neutral points, causing half-cycle saturation and direct current bias in transformer cores, which further lead to increased harmonic currents and reactive power consumption. In severe cases, these effects may result in equipment damage and even large-scale power outages (Hübert et al., 2024; Pratscher et al., 2024). For example, the 1989 Québec blackout in Canada and the 2003 Swedish power grid disturbance were both closely associated with GICs (Blais & Metsa, 1993; Bolduc, 2002; Pulkkinen et al., 2005).

With the continuous expansion of power grid scale and the increase in voltage levels, the potential risks of GICs have gradually extended from high-latitude regions to mid- and low-latitude regions (Gaunt & Coetzee, 2007;

Software: Xuejian Zhang
Supervision: Wenxin Kong
Validation: Xuejian Zhang, Sheng Li
Visualization: Nian Yu
Writing – original draft: Xuejian Zhang
Writing – review & editing:
Wenxin Kong, Zikun Zhou, Nian Yu

Liu et al., 2014). Therefore, forecasting GICs is of great significance for ensuring the safe and stable operation of power systems. However, due to the extreme scarcity of geoelectric field and GIC observation data, directly predicting GICs remains highly challenging (Bailey et al., 2022).

Existing studies usually achieve GIC forecasting by predicting proxy variables of GICs. One category of methods focuses on predicting geomagnetic indices, such as Dst or SYM-H (Collado-Villaverde et al., 2021; Long et al., 2022; Temerin & Li, 2006), although these indices cannot be directly applied to GIC estimation. Another category focuses on the rate of change of geomagnetic disturbances, dB/dt (Lotz & Cilliers, 2015; Priyadarshi, 2024; Wintoft et al., 2015); however, this approach may suffer from low-frequency information loss and noise amplification (Keese et al., 2020; Tóth et al., 2014). In addition, some studies have attempted to directly predict the geomagnetic field B and subsequently calculate the geoelectric field and GICs using subsurface conductivity models. Nevertheless, such methods generally rely on subsurface conductivity structure parameters, thereby introducing additional uncertainties (Keese et al., 2020; Pulkkinen et al., 2013; Upendran et al., 2022; Zhang et al., 2015).

In contrast, the geoelectric field is the direct physical quantity driving GICs (Lotz & Cilliers, 2015), making its prediction more physically reasonable. However, related studies remain limited. For example, Lotz et al. (2017) predicted geoelectric fields using an empirical model, but the electric field data were derived from geomagnetic data and a one-dimensional (1D) conductivity model rather than direct observations. This 1D conductivity model represents the subsurface electrical conductivity structure and takes geomagnetic field disturbances as input for geoelectric field calculation. Bailey et al. (2022) combined LSTM and attention mechanisms to predict geoelectric fields and further estimate GICs; however, the geoelectric fields were still calculated using a 1D conductivity model. In this case, uncertainties in the conductivity model directly affect the accuracy of GIC prediction. Although empirical approaches based on the linear relationship between geoelectric fields and GICs can also be used to estimate GICs (Torta et al., 2012), such methods require large amounts of measured GIC data, which are extremely scarce.

If measured geoelectric fields can be directly predicted, GICs can then be flexibly calculated through either empirical approaches or physics-based models without relying on subsurface conductivity structure parameters, thereby avoiding additional uncertainties introduced by conductivity models and potentially improving the accuracy of GIC prediction. Physics-based models generally construct a direct-current equivalent circuit of the power grid, first calculating transmission line voltages and subsequently computing substation GICs. However, due to the limited availability of measured geoelectric field data, related studies remain scarce.

To address this issue, this study develops a multi-module convolutional neural network combined with a multi-head attention mechanism (MM-CNN-MHA) based on measured geoelectric field data for predicting geoelectric fields and GICs. Taking Fujian Province, China, as the study area, solar wind, geomagnetic disturbance, and geoelectric field observations from 2010 to 2021 are utilized to propose two GIC forecasting approaches: (a) predicting geoelectric fields first and subsequently calculating GICs, and (b) directly predicting GICs. Furthermore, the model results are compared with GIC calculations based on measured geoelectric fields, a 1D conductivity model, and magnetotelluric (MT) impedance data to evaluate the performance differences among these methods.

The remainder of this paper is organized as follows. Section 2 introduces the data sets used in this study. Section 3 presents the models and methodologies. Section 4 describes the model training process. Sections 5 and 6 provide the results and discussion, respectively. Finally, Section 7 summarizes the main conclusions of this study.

2. Data

2.1. Solar Wind Data

We utilized solar wind data from 2010 to 2021, covering nearly the entire 24th solar cycle and extending into the early phase of the 25th cycle. Solar wind parameters measured at the L1 point can provide a lead time of approximately 15–60 min (Bailey et al., 2022).

For missing data, gaps shorter than 15 min were filled using linear interpolation, whereas gaps longer than 15 min were discarded. The interpolation threshold was determined based on both temporal scale considerations and the physical characteristics of the data. On the one hand, this study employed data with a 1-min resolution and used

the preceding 120 min as model input; therefore, a 15-min gap is relatively short and does not significantly disrupt the overall temporal structure of the time series. On the other hand, solar wind parameters and geomagnetic disturbances generally exhibit temporal continuity on minute timescales, allowing short-duration gaps to be reasonably approximated through linear interpolation. However, when the missing duration becomes longer, the data may span different disturbance phases, and simple interpolation may obscure the true variation characteristics and introduce smoothing errors. Therefore, gaps exceeding 15 min were removed from the data set. In addition, this preprocessing strategy is consistent with that adopted in previous studies for solar wind parameter processing (Bailey et al., 2022).

The data were obtained from the high-resolution OMNIWeb data set (1-min resolution, https://omniweb.gsfc.nasa.gov/ow_min.html), which includes solar wind speed, density, temperature, interplanetary magnetic field (IMF), as well as the solar wind dynamic pressure $P_{\text{dyn}} = \rho V_x^2$, where ρ denotes the solar wind mass density, and the y-component of the interplanetary electric field $E_y = V_x B_z$. Among these, the dynamic pressure primarily controls magnetospheric compression and triggers impulsive responses. The y-component of the interplanetary electric field directly regulates magnetospheric energy input by controlling the intensity of magnetic reconnection (Long et al., 2022). During solar wind–magnetosphere interactions, effective dayside magnetic reconnection mainly occurs under southward IMF conditions (i.e., $B_z < 0$), which facilitates the transfer of solar wind energy into the magnetosphere. Consequently, positive E_y values, corresponding to southward IMF conditions, generally represent effective energy input into the magnetosphere, whereas negative E_y values have a relatively weaker contribution to driving geomagnetic disturbances. Based on this physical mechanism and following Long et al. (2022), the rectified electric field E_s was adopted in place of the original E_y . Specifically, only the positive component of E_y was retained, while E_s was set to zero when $E_y < 0$, that is, $E_s = \max(0, E_y)$. This approach considers only the effective solar wind energy input into the Earth's magnetosphere under southward IMF conditions, while neglecting the negative component associated with northward IMF conditions.

Figure 1 presents the solar wind parameters and the IMF during the St. Patrick's Day geomagnetic storm on March 17–18, 2015, including P_{dyn} and E_s . This storm was one of the strongest events in Solar Cycle 24, with a minimum SYM-H index reaching -234 nT. This event was selected as a case study to evaluate model performance under strong geomagnetic disturbance conditions, and its key characteristics are summarized below. As shown in Figure 1, at time t_1 , the sudden increase in solar wind density and P_{dyn} marked the arrival of the interplanetary shock. Subsequently, at t_2 , B_z turned southward, creating favorable conditions for dayside magnetopause reconnection. During the interval from t_2 to t_3 and t_4 to t_5 , the prolonged southward B_z led to a significant enhancement of the rectified electric field E_s , with a peak value of approximately 16 mV/m. The combined effect of high dynamic pressure and sustained southward IMF injected substantial energy into the magnetosphere, driving the development of this intense geomagnetic storm.

2.2. Geomagnetic Observatory Data, Goelectric Field Data, and Geomagnetically Induced Currents Data

Minute-resolution geomagnetic disturbance and goelectric field data simultaneously observed during geomagnetic storms from 2010 to 2021 were provided by the China Earthquake Networks Center, National Earthquake Data Center, at the Quanzhou geomagnetic/geoelectric observatory in Fujian Province, China (24.98°N, 118.59°E). Overall, the data set exhibits high completeness, with missing rates of approximately 0.15% and 0.12% for the geomagnetic disturbance and goelectric field data, respectively.

For the small fraction of missing data, linear interpolation was applied for gap filling. To minimize uncertainties introduced during preprocessing, only short-duration gaps shorter than 15 min were interpolated, whereas gaps exceeding 15 min were removed from the data set.

Outliers were identified using a two-stage detection procedure. First, the Z-score-based 3σ criterion was applied to identify globally extreme outliers, where data points deviating from the mean by more than three standard deviations were classified as anomalous. This procedure identified 9, 9, 13, and 14 global outliers in QZHX, QZHY, EX, and EY, respectively, corresponding to approximately 0.004%, 0.004%, 0.0057%, and 0.0062% of the 226,641 time steps. Second, to detect local anomalies that may not exceed the global 3σ threshold but exhibit unrealistic deviations relative to neighboring observations, a sliding-window Hampel filter was applied. Unlike the Z-score method, the Hampel filter evaluates each data point using the local median and the median absolute deviation (MAD) within the moving window, thereby improving the detection of isolated local anomalies. The

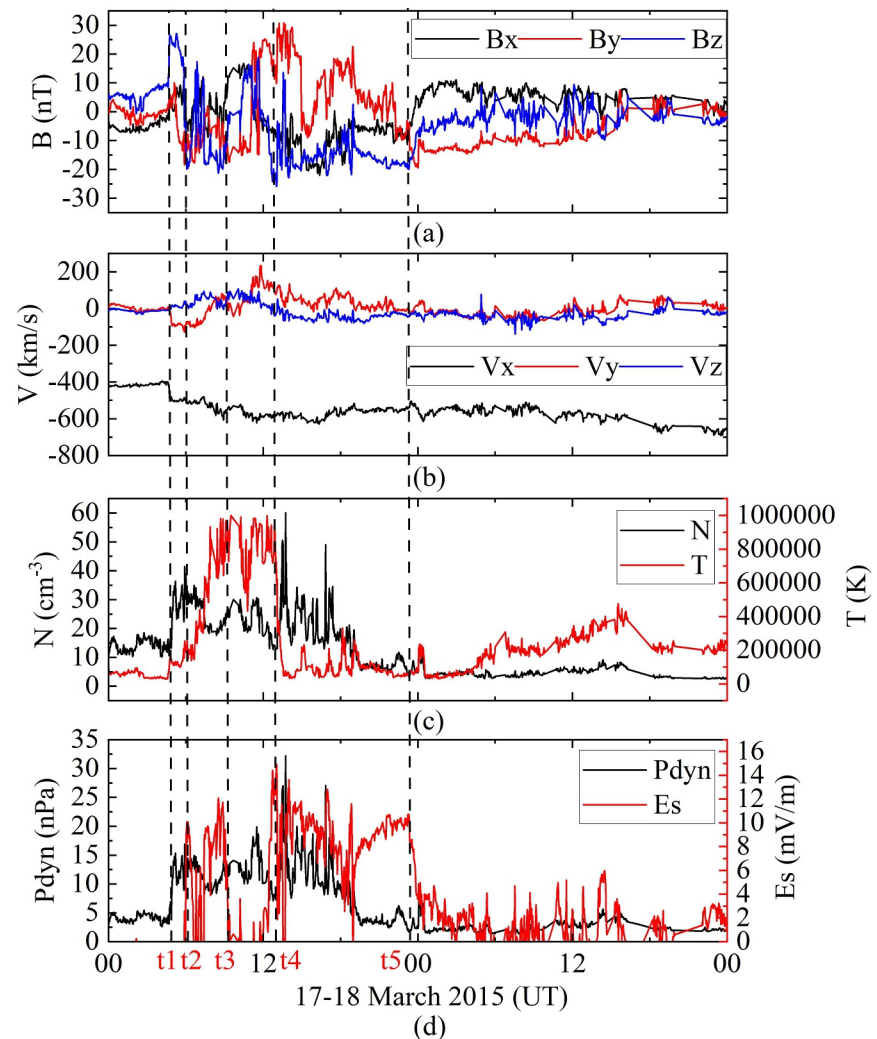


Figure 1. Time series of solar wind parameters, IMF, dynamic pressure, and rectified electric field during March 17–18, 2015. (a) IMF components B_x , B_y , and B_z ; (b) solar wind velocity components V_x , V_y , and V_z ; (c) solar wind density N and temperature T ; (d) dynamic pressure P_{dyn} and rectified electric field E_s .

local filtering procedure additionally identified 10, 12, 16, and 15 anomalous points in QZHX, QZHY, EX, and EY, respectively, accounting for approximately 0.0044%, 0.0053%, 0.0071%, and 0.0066% of the data set. Finally, the identified outliers were corrected using linear interpolation.

We further compared the experimental results obtained with and without the local anomaly filtering procedure and found no noticeable differences between the two cases. This is likely because the number of local anomalies identified in our data set was very small. Nevertheless, the influence of local anomalies should still be considered in future studies, particularly when data sets contain a larger proportion of anomalous observations or have relatively poor data quality, as local anomalies may have a greater impact on model training and prediction performance.

Furthermore, the time derivatives dB_x/dt and dB_y/dt were calculated and cross-correlated with the measured electric fields E_x^{meas} and E_y^{meas} , where x and y denote the north–south and east–west directions, respectively, to remove events with poor correlation.

Geoelectric fields are induced by temporal variations in the geomagnetic field; therefore, under normal conditions, the geoelectric field E^{meas} is expected to exhibit a certain degree of correlation with the geomagnetic field variation rate dB/dt . Low correlation generally indicates the presence of abnormal factors in the data, such as

measurement noise, instrumental errors, or local non-geophysical interference. Including such samples directly in the training dataset may introduce substantial label noise, thereby degrading both the predictive accuracy and physical consistency of deep learning models.

Following Bailey et al. (2022), observed geoelectric field values exceeding 200 mV/km were truncated to 200 mV/km to mitigate the adverse influence of extreme values on model training. It should be noted that when outliers constitute a substantial fraction of the data set, truncation or removal may improve statistical evaluation metrics while simultaneously reducing the model's capability to reproduce realistic extreme events. Therefore, such preprocessing strategies should be applied with caution.

In the present study, after removing the identified anomalous samples, the proportion of extreme samples was extremely small. Among a total of 226,641 time steps, only two samples exceeded 200 mV/km, accounting for approximately 0.0009% of the data set.

Future studies should further evaluate the impacts of different outlier-processing strategies on both model performance and practical applicability under different data distribution characteristics.

Zhang et al. (2020) provided GIC measurements at the Huangmeishan (HMS) 220-kV substation (27.37°N, 119.53°E) in Fujian Province during the geomagnetic storms of March 17–18 and December 14–15, 2015. In the present study, the 17–18 March 2015 event was selected for detailed analysis because of its stronger geomagnetic intensity. Figure 2 shows the measured geomagnetic disturbances and their horizontal magnetic field variation rate at the Quanzhou observatory, the measured geoelectric field at the same site, and the observed GICs at the HMS substation.

The horizontal geomagnetic field variation rate, dB_H/dt , is defined as the combined magnitude of the temporal variation rates of the two orthogonal horizontal geomagnetic disturbance components, that is, $dB_H/dt = \sqrt{(dB_x/dt)^2 + (dB_y/dt)^2}$, where B_x and B_y denote the two orthogonal horizontal geomagnetic disturbance components, respectively.

It can be observed that the variation rate of geomagnetic disturbances is highly correlated with the amplitudes of the measured electric field and GICs: in general, the faster the geomagnetic disturbance varies, the larger the amplitudes of the measured electric field and GICs become. Moreover, the measured electric field and GICs exhibit a similar variation trend. This indicates that the variation rate of geomagnetic disturbances has a significant impact on both the electric field and GICs. It should be noted that the distance between the Quanzhou geomagnetic station and the HMS substation is approximately 280 km, which meets the requirements for GIC studies, as justified in Zhang et al. (2020).

2.3. Surface Wave Impedance Data and Power Grid Parameters

In the subsequent experiments, the predicted GICs need to be compared with three physics-based modeling results, which are calculated based on a 1D conductivity model, MT-measured impedance data, and measured geoelectric field data, respectively. Therefore, it is necessary to introduce the conductivity model, MT impedance data, and power grid parameters used in this study.

Table 1 presents the 1D conductivity model for Fujian Province adopted from Zhang et al. (2020). This 1D model can be regarded as a regional equivalent approximation of the complex three-dimensional subsurface electrical structure in Fujian. The corresponding resistivities are 200 $\Omega\cdot\text{m}$ for depths of 0–30 km, 76.9 $\Omega\cdot\text{m}$ for 30–90 km, 25 $\Omega\cdot\text{m}$ for 90–150 km, and 3.33 $\Omega\cdot\text{m}$ for depths greater than 150 km. Three-dimensional MT inversion results for southeastern South China reported by Xu et al. (2019) indicate that the subsurface electrical structure in this region exhibits pronounced depth-dependent variations. Their results suggest that shallow crustal fault zones, sedimentary cover, and conductive middle-crustal layers all contribute significantly to the regional electrical structure. Therefore, the resistivity value of 200 $\Omega\cdot\text{m}$ adopted for the 0–30 km layer in this study can be regarded as an equivalent representation of the combined effects of multiple shallow electrical units.

For the 30–90 km depth range, Xu et al. (2019) reported lithospheric mantle resistivities of approximately 30–100 $\Omega\cdot\text{m}$, which are generally consistent in magnitude with the value of 76.9 $\Omega\cdot\text{m}$ used in this study. The resistivity value of 25 $\Omega\cdot\text{m}$ for the 90–150 km layer reflects enhanced conductivity within the deeper lithospheric

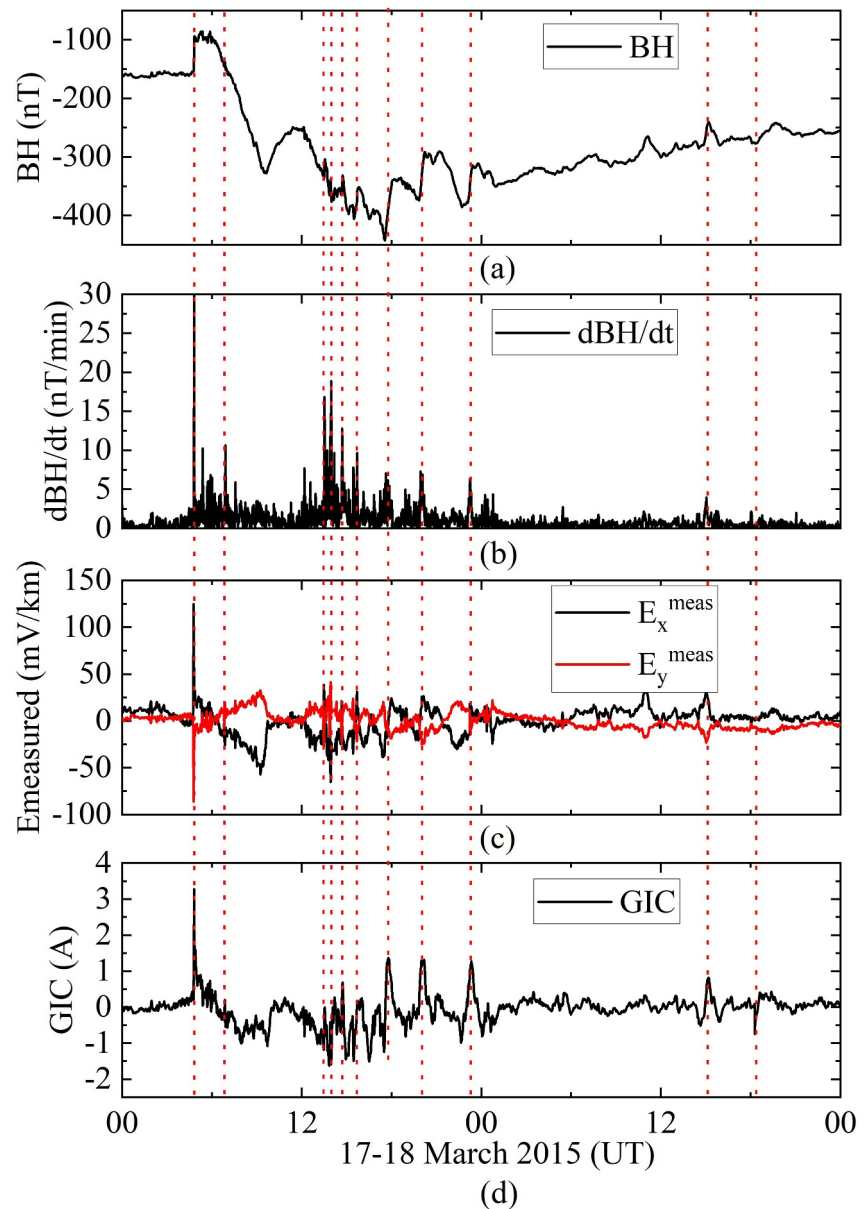


Figure 2. Time series of measured geomagnetic data and their variation rates, and the measured geoelectric field at the Quanzhou observatory, as well as the measured GICs at the HMS substation during the geomagnetic storm of March 17–18, 2015. (a) Horizontal components of the geomagnetic data; (b) Variation rates of the horizontal geomagnetic components; (c) Measured geoelectric field components E_x^{meas} and E_y^{meas} at Quanzhou; (d) Measured GICs at the HMS substation.

Table 1
1D Conductivity and Resistivity Model of Fujian Province

Depth (km)	Conductivity (S/m)	Resistivity ($\Omega \cdot \text{m}$)
0–30	0.005	200
30–90	0.013	76.9
90–150	0.04	25
>150	0.3	3.33

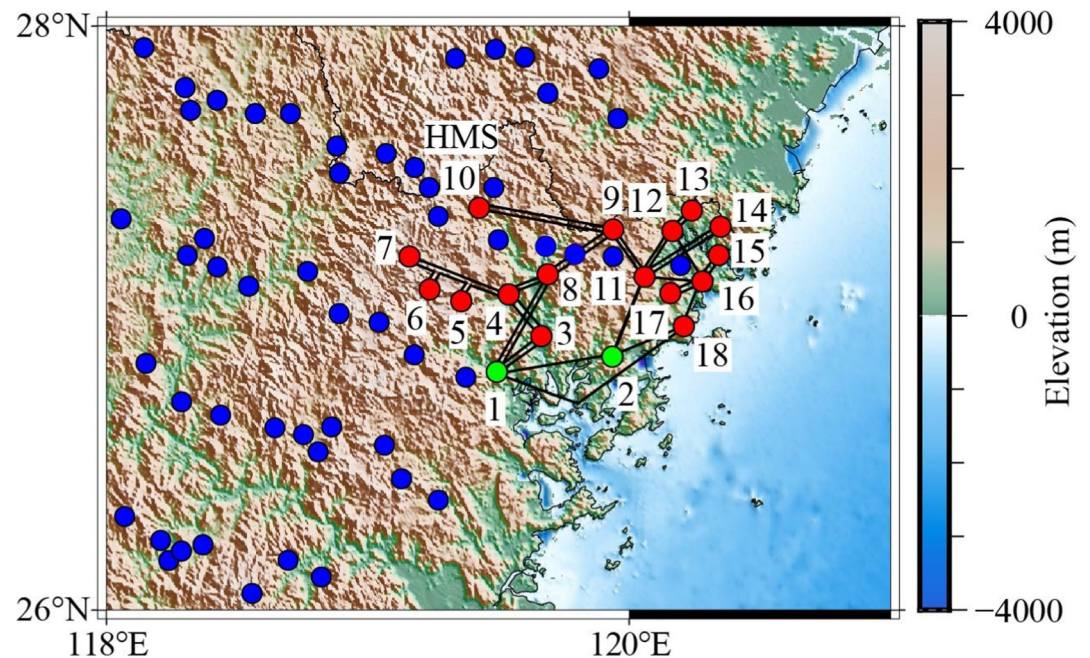


Figure 3. Topology of the 220 kV transmission network in Ningde, Fujian Province, and distribution of MT measurement sites. Blue dots represent MT measurement sites, green dots indicate 500 kV substations, and red dots indicate 220 kV substations. Numbers correspond to substation identifiers, with number 10 representing the HMS 220 kV substation. Double lines denote double-circuit lines, and single lines denote single-circuit lines.

mantle and the lithosphere–asthenosphere transition zone. The value of $3.33 \Omega\cdot\text{m}$ below 150 km mainly represents the deep conductive background structure.

In addition, MT-measured impedance data for Fujian Province were provided by Xu et al. (2019) (see Figure 3). Relevant information on the 220 kV power grid in Ningde, Fujian, was obtained from the State Grid Corporation of China. Due to confidentiality requirements, specific substation names are not shown in the figure and are represented by numbers, with number 10 corresponding to the HMS 220 kV substation. The surrounding network of this substation mainly consists of 220 kV lines. Although two 500 kV substations are depicted in the figure, they are not included in the modeling calculations.

These power grid data are sufficient for calculation purposes because the GIC at a given substation is primarily influenced by the line parameters of neighboring substations and the regional Earth conductivity structure (Pirjola, 2005).

3. Methodology and Models

3.1. Physics-Based GIC Calculation

Two categories of methods were employed in this study to calculate or estimate GICs. The first category consists of physics-based approaches based on direct-current equivalent circuits of the power grid and geoelectric field integration. In these approaches, geoelectric fields are first used to calculate induced voltages along transmission lines through geoelectric field integration, after which GICs at substations are calculated using the direct-current equivalent circuit of the power grid. The second category involves linear fitting approaches based on the empirical relationship between measured geoelectric fields and observed GICs.

It should be emphasized that the physics-based methods described in this section were not used to generate training labels for the direct GICs prediction model. Instead, they were mainly used to calculate GICs driven by the 1D conductivity model, MT impedance data, and measured geoelectric fields, which serve as physics-based benchmark results for subsequent comparison with deep learning predictions.

The geoelectric field is generally determined by both geomagnetic disturbances and the Earth's conductivity structure. Its frequency-domain representation is given by (Kelbert & Lucas, 2020):

$$\mathbf{E}_{\text{GEO}} = \mathbf{Z} \cdot \mathbf{H} \quad (1)$$

where $\mathbf{Z}$ is the impedance tensor (Ω), $\mathbf{E}_{\text{GEO}}$ represents the geoelectric field (V/m), and $\mathbf{H}$ denotes the magnetic field (A/m).

For a three-dimensional Earth conductivity structure, all four components of the impedance tensor $\mathbf{Z}$ are non-zero complex numbers, expressed as (Chen et al., 2025; Cordell et al., 2021):

$$\mathbf{Z}(f, x, y) = \begin{bmatrix} Z_{xx}^f & Z_{xy}^f \\ Z_{yx}^f & Z_{yy}^f \end{bmatrix} \quad (2)$$

For a 1D medium, the diagonal elements of the impedance matrix are zero, and the off-diagonal elements are mutually opposite (Cordell et al., 2021):

$$\mathbf{Z}(f, x, y) = \begin{bmatrix} 0 & Z_{xy}^f \\ Z_{yx}^f & 0 \end{bmatrix}, \quad (3)$$

where $Z_{yx} = -Z_{xy}$.

The calculated geoelectric field is then interpolated along the transmission lines and integrated to obtain the line voltage (Boteler & Pirjola, 2017; Pirjola, 2002):

$$U_{ab} = \int_a^b \overrightarrow{E}_{\text{GEO}} \cdot d\vec{l}, \quad (4)$$

where U_{ab} denotes the voltage, and $\overrightarrow{E}_{\text{GEO}}$ represents the geoelectric field, and a and b are the endpoints of the transmission line.

Finally, GICs can be calculated using the nodal admittance method (Lehtinen & Pirjola, 1985):

$$\mathbf{I} = (\mathbf{1} + \mathbf{Y}\mathbf{Z})^{-1} \mathbf{J}, \quad (5)$$

where $\mathbf{I}$ represents the current at the transformer, $\mathbf{1}$ is the identity matrix, $\mathbf{Y}$ is the admittance matrix, $\mathbf{Z}$ is the impedance tensor, and $\mathbf{J}$ is the matrix of currents flowing into the ground at all nodes.

The total of all currents I_{ab} flowing into node b from the power grid is defined as J_b . According to Ohm's law:

$$J_b = \sum_{a \neq b} I_{ab} = \sum_{a \neq b} \frac{U_{ab}}{R_{ab}}, \quad (6)$$

where R_{ab} denotes the equivalent resistance between two nodes.

3.2. GIC Calculation Using Linear Fitting

Unlike the physics-based approach described in Section 3.1, the linear fitting method introduced in this section is primarily used to construct training targets for the direct GIC prediction model. Specifically, continuous observed GIC data are unavailable for most geomagnetic storm events, while generating large amounts of training samples using physics-based modeling is computationally expensive. Therefore, this study estimates GICs using the empirical linear relationship between the measured geoelectric fields E_x^{meas} , E_y^{meas} , and observed GICs, and

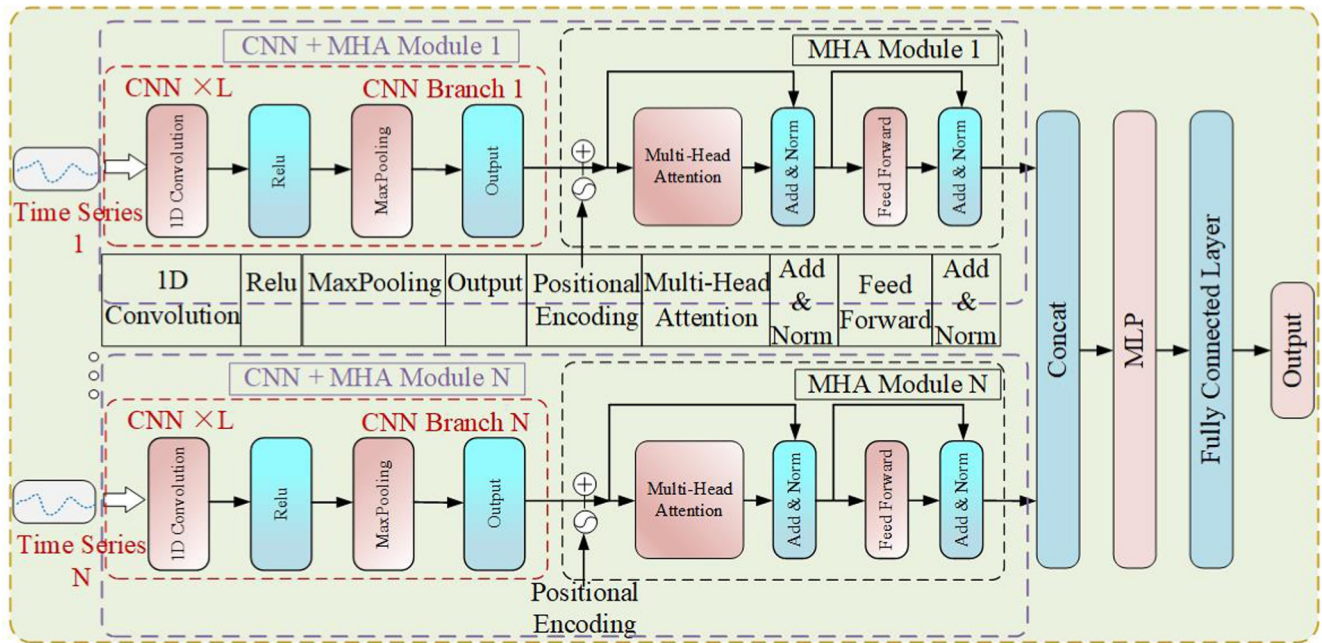


Figure 4. Overall structure of MM-CNN-MHA model.

subsequently uses the estimated GICs as training labels for the MM-CNN-MHA-GIC model. The specific formula is as follows (Bailey et al., 2022):

$$\text{GIC} = a \cdot E_x^{\text{meas}} + b \cdot E_y^{\text{meas}}, \quad (7)$$

where a and b are real coefficients corresponding to a specific substation, with units of A-km/V.

Overall, the linear fitting method has both advantages and limitations. One important advantage is its high computational efficiency. In addition, the empirical coefficients a and b can implicitly capture site-specific characteristics of the actual power-grid response, including the effects of the local network layout, transmission line orientation, grounding conditions, transformer connections, and other operational characteristics, some of which may not be fully represented in the simplified physics-based grid model (Bailey et al., 2022; Zhang et al., 2020). However, the fitted empirical coefficients are substation-specific and rely on the availability of observed GICs. Moreover, changes in the operational grid configuration, grounding conditions, or transformer connections may alter the relationship between the geoelectric field and GICs, thereby reducing the applicability of fixed empirical coefficients. Likewise, if the network topology and electrical parameters used in the physics-based nodal admittance method remain unchanged, the method's prediction accuracy will also be unaffected by such changes. Therefore, when substantial changes occur in the power-grid system, the empirical coefficients or the physics-based grid model should be updated accordingly to ensure accurate GIC calculations.

3.3. Multi-Module CNN - MHA Model

In the field of GIC prediction, various deep learning methods have been applied to forecast GICs or their proxies, including artificial neural networks (ANN) (Keese et al., 2020), recurrent neural networks (RNN) (Priyadarshi, 2024), gated recurrent units (GRU) (Upendran et al., 2022), and long short-term memory networks (LSTM) (Zainuddin et al., 2024). In recent studies, Bailey et al. (2022) found that combining LSTM with a basic attention mechanism for geoelectric field prediction outperforms most other deep learning models. However, due to differences in prediction targets, training data, and other factors, the predictive performance of different models may still vary.

We develop a multi-module convolutional neural network combined with a multi-head attention mechanism (MM-CNN-MHA) model (See Figure 4), in which the MHA module is used to enhance the modeling of global

dependencies in time series. Although LSTM is effective in modeling temporal dependencies, its performance may degrade for long input sequences in practice due to accumulated gradient effects and limited parallelization. In contrast, MHA computes attention weights between each position and all other positions in the sequence, enabling global information interaction (see the formulas in Vaswani et al. (2017)). This enables the modeling of long-range temporal dependencies through direct interactions between different time steps, alleviating some of the limitations encountered by recurrent architectures when handling long sequences.

Nevertheless, MHA has limited capability in capturing local features. Considering that geomagnetic disturbances and solar wind parameters may exhibit strong short-term correlations, such as local abrupt changes or peak events, these local patterns are critical for predicting transient responses of geoelectric fields and GICs. To address this limitation, a CNN is incorporated into the model to automatically extract local features. In addition, the input variables originate from heterogeneous physical processes, including solar wind forcing, magnetospheric responses, and ground induction effects, each of which exhibits distinct statistical characteristics. To account for this heterogeneity and reduce cross-interference among features, a multi-module architecture is adopted, in which physically related inputs are grouped and processed by separate CNN–MHA modules. After each CNN–MHA module extracts features, all the features are concatenated together. These features are then fused through a multi-layer perceptron, and the final prediction is generated via a fully connected layer.

3.3.1. Data Preparation

In this study, the selected input features include solar wind parameters (speed, temperature, density), the dynamic pressure, the rectified electric field, the IMF, and measured geomagnetic disturbances at the Quanzhou station. We include geomagnetic disturbance data as input features. This does not cause information leakage, because the geoelectric field data used in this study are obtained from direct measurements rather than computed from geomagnetic disturbance data. If the geoelectric field were calculated using geomagnetic disturbances and a 1D conductivity model (Bailey et al., 2022), the theoretical geoelectric field would already contain all the geomagnetic disturbance information. In such a case, including geomagnetic disturbances as additional inputs could cause information leakage, resulting in unrealistically idealized predictions. Moreover, the generation of the geoelectric field is directly driven by geomagnetic disturbances. Including geomagnetic disturbances as inputs helps the model learn the true relationship between the two.

Because the overall amplitudes of the geoelectric field in the data set were generally low, high-amplitude disturbance events were relatively rare. If all samples were directly used for model training, the model would tend to bias toward low-amplitude outputs, thereby reducing its capability to predict high-amplitude events. To alleviate this issue, a selective sampling strategy following Bailey et al. (2022) was adopted for training data preparation. When the geoelectric field E^{meas} was within the range of 0–50 mV/km, the samples were under-sampled. For higher-amplitude samples exceeding this range, data augmentation was applied by duplicating samples and introducing random temporal shifts ranging from –10 to +10 min in the input data. In addition, to reduce temporal dependence between neighboring samples, all training samples were separated by at least 60 min.

After data processing, each feature contains 226,641 valid samples. The data were split chronologically to preserve temporal continuity and avoid information leakage, and were divided into training (80%), validation (10%), and test (10%) sets. In addition, the geomagnetic storms on 17–18 March and 14–15 December 2015 were used for event-based evaluation because observed GIC data are available at the HMS substation for these periods. However, because the March 2015 event was also used to calibrate the empirical coefficients in the linear-fitting method, it should not be regarded as a fully independent validation event for the linear-fitting-based GIC estimation or the MM-CNN-MHA-GIC framework. By contrast, the December 2015 event provides an independent evaluation case for the fitted empirical relationship.

According to the physical roles of different input features in the solar wind–magnetosphere–ground induction coupling process, the input parameters were divided into five groups. The first group consists of interplanetary magnetic field (IMF) parameters, including B_x , B_y , and B_z . The second group includes solar wind velocity, dynamic pressure (P_{dyn}), and the rectified electric field (E_s). The third group contains geomagnetic disturbance parameters (QZHX and QZHY). The fourth and fifth groups correspond to solar wind density (N) and solar wind temperature (T), respectively.

Table 2

Hyperparameters for MM-CNN-MHA-E Training

	Embedding dimension	Number of attention heads	Dropout rate	Number of MHA layers	Number of CNN layers
MM-CNN-MHA-Ex	128	4	0.1	1	1
MM-CNN-MHA-Ey	128	4	0.1	1	1

Note. The hyperparameters used for training two different MM-CNN-MHA models: one for E_x^{meas} and one for E_y^{meas} .

The IMF parameters mainly characterize magnetic reconnection conditions between the solar wind and the magnetosphere and were therefore assigned to the same module. Solar wind velocity, dynamic pressure, and rectified electric field primarily describe the dynamical driving of the magnetosphere by the solar wind and were thus grouped. In contrast, geomagnetic disturbance parameters represent the ground response of the magnetosphere–ionosphere system and are more directly related to geoelectric fields and GICs; therefore, they were modeled independently. Solar wind density mainly affects magnetospheric compression processes, whereas solar wind temperature primarily reflects the thermodynamic state of the solar wind plasma. Accordingly, these two parameters were treated as separate input modules.

In addition, correlations among the input features were further analyzed to investigate their potential impacts on feature grouping and prediction performance. As shown in Table S1 of Supporting Information S1, a strong correlation exists between B_z and E_s (-0.870), while solar wind density N and dynamic pressure P_{dyn} also exhibit a relatively strong correlation (0.822). However, the feature grouping strategy in this study was not determined solely based on statistical correlations but was instead primarily guided by the underlying physical processes represented by each variable. For example, E_s was assigned to the solar wind dynamical driving module rather than the IMF module, whereas P_{dyn} was grouped with solar wind velocity to characterize the dynamical forcing of the magnetosphere by the solar wind.

To further evaluate the influence of feature grouping, a second feature grouping scheme based on correlation relationships was also constructed, in which E_s was reassigned to the IMF module and P_{dyn} was grouped with N . Under otherwise identical experimental conditions, the two grouping schemes produced overall similar prediction results for E_x^{meas} and E_y^{meas} , indicating that feature correlations have a relatively limited impact on model prediction performance.

Each group of features is fed into its corresponding CNN–MHA module for training. The model uses data from the 120 min preceding the prediction time as input to forecast geoelectric fields and GICs for the following 60 min. The prediction horizon is set to 60 min to provide power grid companies with sufficient response time; in contrast, most existing studies use prediction horizons of less than 60 min.

4. Training the MM-CNN-MHA Model

We use the MM-CNN–MHA model to predict GICs through two complementary approaches: (a) predicting the geoelectric field and subsequently calculating GICs at the HMS substation, and (b) directly predicting GICs using proxy GIC values derived from linear fitting of the measured geoelectric field data.

4.1. Predicting Geoelectric Fields E_x^{meas} and E_y^{meas}

This approach for predicting geoelectric fields is referred to as MM-CNN–MHA–E. A grid search strategy was employed in this study to systematically optimize the model hyperparameters within a predefined hyperparameter space. Specifically, combinations of key hyperparameters, including the embedding dimension (d_{model}), number of attention heads (n_{head}), dropout rate, number of multi-head attention (MHA) layers, feedforward network dimension, number of CNN layers, and multilayer perceptron (MLP) structure, were explored. In total, 2592 valid hyperparameter combinations were generated and individually evaluated through model training and validation.

Each hyperparameter configuration was trained using the training data set and evaluated based on the RMSE metric on the validation data set. The final hyperparameter set was selected according to its overall performance within the predefined search range as well as the relative stability of the training process.

Table 3
Hyperparameters for MM-CNN-MHA-GIC Training

	Embedding dimension	Number of attention heads	Dropout rate	Number of MHA layers	Number of CNN layers
MM-CNN-MHA-GIC	128	4	0.1	1	1

The hyperparameter settings are listed in Table 2. The model uses the Huber loss function, which behaves like mean squared error (MSE) for small errors and like mean absolute error (MAE) for large errors, balancing robustness to storm-time outliers with sensitivity to small-to-moderate errors. The model is trained using the Adam optimizer, and the Rectified Linear Unit (ReLU) is employed as the activation function.

4.2. Predicting GICs

We also evaluate a direct GIC prediction approach, referred to as MM-CNN-MHA-GIC, in which proxy GIC values derived from the measured geoelectric field are used as training targets. The hyperparameters used for model training are listed in Table 3.

Similar to Bailey et al. (2022), we first use the linear relationship between the geoelectric field and measured GICs to fit the empirical coefficients a and b via regression (see Equation 7). Once a and b are obtained, Equation 7 is used to calculate GICs during other geomagnetic storms for model training, and to ultimately predict GICs at the HMS substation. This approach effectively reduces computational cost. Although it was initially based on the plane-wave assumption, previous studies have shown that measured geoelectric fields can also be used for GIC calculations when the spatial distribution of the geoelectric field is relatively uniform (Pulkkinen et al., 2007). Furthermore, Zhang et al. (2020) demonstrated that, despite the distance of approximately 280 km between the Quanzhou geomagnetic station and the HMS substation, Equation 7 can still be used to estimate substation GIC.

The measured GICs at the HMS substation during March 17–18, 2015, was first fitted against the simultaneously measured geoelectric fields at the Quanzhou station, yielding empirical coefficients of $a = 6.39$ A-km/V and $b = -19.24$ A-km/V.

As shown in Figure 5, the linearly fitted GICs and the GICs calculated using the physics-based modeling framework show good agreement with the observed GICs in both amplitude and temporal variation. Here, the physics-based modeling framework refers to the GIC calculation method described in Section 3.1, in which the geoelectric field is first integrated along transmission lines to obtain induced line voltages, and the resulting GICs are subsequently calculated using the nodal admittance method based on the direct-current equivalent circuit of the power grid. Specifically, the GICs obtained by applying the measured geoelectric field, E^{meas} that is, the measured geoelectric field components E_x^{meas} and E_y^{meas} , to this framework are hereafter referred to as E^{meas} -Physics-Based GIC.

For example, at 04:48 on 17 March 2015, the observed GIC reached a peak of 3.28 A, while the E^{meas} -Physics-Based GIC and the Linearly Fitted GIC were 3.60 and 2.46 A, respectively. Statistical metrics indicate that the Linearly Fitted GIC achieved a Pearson correlation coefficient (PCC) of 0.76, a root mean square error (RMSE) of 0.26 A, and a mean error (ME) of 0.03 A with respect to the observed GIC. In comparison, the E^{meas} -Physics-Based GIC achieved a PCC of 0.60, an RMSE of 0.28 A, and an ME of 0.06 A.

These results indicate that the linear fitting method slightly outperforms physics-based modeling, which is consistent with previous studies (Bailey et al., 2022). The relatively low PCC may be attributed to measurement noise and the considerable distance between the geoelectric station and the substation. Nevertheless, the accuracy of physics-based modeling is sufficient for engineering applications, validating the reliability of the GIC model for the 220 kV power grid in Ningde, Fujian Province.

It should be noted that the empirical coefficients, a and b , used in the linear fitting method were calibrated using the measured geoelectric field and observed GIC data during the 17–18 March 2015 geomagnetic storm.

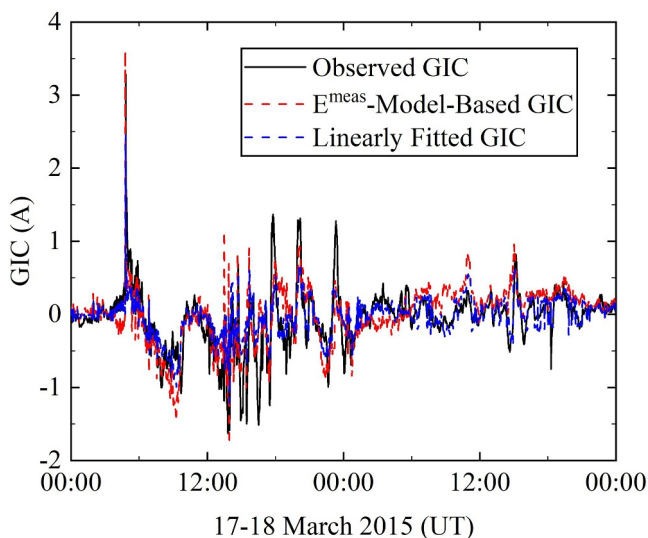


Figure 5. GICs time series for HMS substation from March 17–18, 2015.

Therefore, the agreement between the linearly fitted GICs and the observed GICs for this event should be regarded as an in-sample evaluation rather than a strict out-of-sample validation. The primary purpose of introducing this empirical relationship was not to establish an independent GIC prediction model, but to characterize the local relationship between the measured geoelectric field and GIC at the HMS substation and to generate the proxy GIC data used to train the MM-CNN-MHA-GIC model.

Furthermore, applying the empirical coefficients a and b to the geomagnetic storm on 14–15 December 2015 yields a PCC of 0.72, an RMSE of 0.18, and an ME of 0.03 between the Linearly Fitted GIC and the observed GIC, indicating a small positive bias and confirming the applicability of the proposed method in this study.

Therefore, the GICs at the HMS substation obtained via linear fitting during various geomagnetic storms are used as training data, and the MM-CNN-MHA-GIC model is ultimately used to predict the observed GIC for the storms on March 17–18 and December 14–15, 2015. It should be noted that, because the empirical coefficients a and b were calibrated using the 17–18 March 2015 geomagnetic storm, and this event was subsequently used for model performance evaluation, a certain degree of potential data leakage may exist for the March 2015 event. Therefore, this event should not be regarded as a fully independent validation case for the MM-CNN-MHA-GIC model. To further assess whether this potential data leakage affects the main conclusions of this study, we additionally evaluated the model performance using the independent 14–15 December 2015 geomagnetic storm in Section 5.4. The results show that the model performance ranking obtained from the December 2015 event is consistent with that obtained from the combined March and December 2015 events, suggesting that the potential data leakage associated with the March 2015 event has a limited impact on the main scientific conclusions of this study. However, owing to the limited availability of observed GIC data in Fujian Province, only the 17–18 March and 14–15 December 2015 geomagnetic storms could be used for validation in this study. This is a limitation of the present work. Future studies should include more independent geomagnetic storm events with observed GIC records and should strictly separate empirical-coefficient calibration, model training, and model testing to further evaluate the generalization ability and applicability of the proposed method.

4.3. Evaluating the Model Skill

We use two types of model evaluation metrics to assess predictive performance: point-wise error metrics and event-based metrics. In addition to the RMSE, PCC, and ME, the point-wise error metrics include the normalized root mean square error (nRMS). These metrics evaluate the consistency between predictions and observations in both trend and amplitude, allowing an accurate assessment of how closely the predictions match reality. The nRMS is defined as follows:

$$nRMS(y, \hat{y}) = \sqrt{\frac{\langle (y - \hat{y})^2 \rangle}{\langle y^2 \rangle}}, \quad (8)$$

where y denotes the observed values, $\hat{y}$ denotes the predicted values, and $\langle \dots \rangle$ represents the arithmetic mean. An nRMS of 0 indicates perfect agreement between the model and observations; Values of nRMS < 1 generally indicate reasonable agreement, whereas nRMS > 1 suggests that the prediction errors are comparable to or larger than the mean observed magnitude.

One important purpose of introducing nRMS in this study is to facilitate direct comparison with previous work, particularly the GIC simulations of Zhang et al. (2020) based on the Space Weather Modeling Framework (SWMF) (see Section 5.4, Comparison of Predicted GIC and GIC Calculated Using 1D Conductivity, MT Impedance, and Measured Geoelectric Field During the March and December 2015 Geomagnetic Storms). However, nRMS also has certain limitations. Geomagnetic field, geoelectric field, and GIC time series are typically highly imbalanced, with the vast majority of samples corresponding to quiet periods and only a small proportion representing geomagnetic storm disturbances. Under these conditions, the normalization term $\sqrt{\langle y^2 \rangle}$ is dominated by the numerous near-zero observations during quiet periods, which may reduce the sensitivity of nRMS to differences in model performance during disturbed periods, especially when evaluating the ability to capture extreme peaks. For example, two models may exhibit similar performance during quiet periods but substantially different peak prediction capabilities during geomagnetic storms. Since the normalization term

depends only on the observed time series and is dominated by quiet-period observations, the resulting nRMS values may differ only slightly, making it difficult to fully reflect differences in model performance during extreme disturbance events. Therefore, nRMS alone cannot comprehensively characterize model performance during geomagnetic storms. To obtain a more comprehensive and reliable evaluation, this study complements nRMS with point-wise error metrics (RMSE and ME) and event-based metrics (POD, POFD, and HSS), providing a comprehensive assessment of model performance in terms of prediction error, trend consistency, and event detection capability.

Event-based evaluation metrics follow Pulkkinen et al. (2013). The threshold is selected based on the statistical distribution of the data. Based on the event determination from predictions and observations, four outcomes can be obtained: correctly identified events (true positives, TP), missed events (false negatives, FN), falsely identified events (false positives, FP), and correctly identified non-events (true negatives, TN).

From these outcomes, various evaluation metrics can be derived. The probability of detection (POD) represents the likelihood that the model correctly predicts events and is defined as follows:

$$\text{POD} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (9)$$

The probability of false detection (POFD) represents the likelihood of falsely predicting an event and is defined as follows:

$$\text{POFD} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (10)$$

Finally, a comprehensive metric, the Heidke Skill Score (HSS), is used to measure the improvement of the model's predictive accuracy relative to random prediction. An HSS < 0 indicates that the model performs worse than random prediction; HSS = 0 indicates performance equivalent to random prediction; and HSS = 1 indicates perfect predictive skill. HSS is calculated as follows:

$$\text{HSS} = \frac{2 \times (\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN})}{(\text{TP} + \text{FN})(\text{FN} + \text{TN}) + (\text{TP} + \text{FP})(\text{FP} + \text{TN})} \quad (11)$$

In addition, a persistence-based empirical model is used as a benchmark for the MM-CNN-MHA model. This benchmark assumes that the predicted values for the next 60 min are equal to the current values. The proposed MM-CNN-MHA model is expected to outperform the baseline model, which serves as a reference for assessing model skill in most metrics.

5. Results

5.1. Comparison to Existing Methods

Following the methodology of Upendran et al. (2022), systematic comparisons were conducted on the 10% testing data set using both point-wise evaluation metrics, including RMSE, ME, and nRMS, and event-based evaluation metrics. The performance of the MM-CNN-MHA-E model in predicting E_x^{meas} and E_y^{meas} was compared with that of the single-module CNN-MHA-E (SM-CNN-MHA-E), ANN, RNN, GRU, LSTM, and attention-based LSTM models.

All comparison models were implemented and trained using the same data set, identical input features, and the same data partitioning scheme (training, validation, and testing data sets) as the MM-CNN-MHA model to ensure fairness and consistency in the comparisons.

Tables 4 and 5 summarize the point-wise performance metrics of all models for E_x^{meas} and E_y^{meas} prediction, respectively. The results show that the MM-CNN-MHA-E model achieved the lowest RMSE and nRMS values for both E_x^{meas} and E_y^{meas} , indicating the highest overall prediction accuracy among all tested models. This suggests that grouping input features during model training contributes to improved predictive performance. In terms of systematic bias, the absolute ME values of the MM-CNN-MHA-E model were generally smaller than

Table 4
RMSE, ME, and nRMS of Different Deep Learning Models for Predicting E_x^{meas}

	MM-CNN-MHA-E	SM-CNN-MHA-E	ANN	RNN	GRU	LSTM	LSTM + attention
RMSE (mV/km)	20	21	24	30	30	24	22
ME (mV/km)	1	1	−10	1	8	−3	−8
nRMS	0.65	0.67	0.79	0.9	0.9	0.77	0.72

those of most baseline models, indicating that the model maintained relatively low systematic bias while achieving high prediction accuracy.

In addition, the PCC values of the predicted E_x^{meas} and E_y^{meas} from the MM-CNN-MHA-E model were 0.65 and 0.68, respectively. By comparison, Bailey et al. (2022) reported PCC values of 0.60 and 0.61, whereas Honkonen et al. (2018) reported PCC values of 0.62 and 0.25, respectively.

In addition, event-based evaluation metrics were further analyzed in this study. For E_x^{meas} , threshold values of 30, 40, 50, 60, and 70 mV/km were adopted, whereas thresholds of 10, 20, 30, 40, and 50 mV/km were used for E_y^{meas} , corresponding to geomagnetic disturbance events ranging from weak to strong intensities.

As shown in Tables S2 and S3 of Supporting Information S1, both POD and HSS generally decrease with increasing event thresholds for all models, indicating that high-amplitude geoelectric field events are more difficult to predict. Overall, the MM-CNN-MHA-E model achieved relatively high HSS values under most threshold conditions and maintained a favorable balance between high probability of detection (POD) and low probability of false detection (POFD).

Furthermore, we compared the performance of the MM-CNN-MHA-GIC model with that of the single-module CNN-MHA-GIC (SM-CNN-MHA-GIC), ANN, RNN, GRU, LSTM, and LSTM with attention mechanism models in predicting GICs on the same 10% test set.

As shown in Table 6, in terms of point-wise evaluation metrics, the MM-CNN-MHA-GIC model achieved the lowest RMSE (0.27 A) and the smallest nRMS (0.68), indicating the highest overall prediction accuracy among the tested models. In terms of systematic bias, the corresponding ME value was −0.07 A, indicating relatively small overall bias and performance comparable to that of most baseline models.

In addition, the PCC value of the predicted GICs in this study was 0.61, which is close to the value of 0.60 reported by Bailey et al. (2022), further supporting the reliability of the model results.

In addition, event-based evaluations were further conducted for the predicted GICs. Threshold values of 0.2, 0.4, 0.6, 0.8, and 1 A were adopted in this study, approximately corresponding to geoelectric field events with different intensity levels.

As shown in Table S4 of Supporting Information S1, both POD and HSS generally decrease with increasing event thresholds for all models, indicating that high-amplitude GIC events are more difficult to predict. Overall, the MM-CNN-MHA-GIC model achieved relatively high HSS values under most threshold conditions and maintained a favorable balance between event detection capability and false alarm rate.

Furthermore, the overall capability of the MM-CNN-MHA-GIC model for direct GIC prediction was generally comparable to that of the MM-CNN-MHA-E model for geoelectric field prediction. This result is broadly consistent with the conclusions reported by Bailey et al. (2022).

Table 5
RMSE, ME, and nRMS of Different Deep Learning Models for Predicting E_y^{meas}

	MM-CNN-MHA-E	SM-CNN-MHA-E	ANN	RNN	GRU	LSTM	LSTM + attention
RMSE (mV/km)	18	19	25	22	24	20	21
ME (mV/km)	−2	−2	9	4	4	3	−4
nRMS	0.68	0.72	0.90	0.88	0.90	0.75	0.83

Table 6
RMSE, ME, and nRMS of Different Deep Learning Models for Predicting GICs

	MM-CNN-MHA-GIC	SM-CNN-MHA-GIC	ANN	RNN	GRU	LSTM	LSTM + attention
RMSE (A)	0.27	0.29	0.34	0.31	0.31	0.30	0.38
ME (A)	−0.07	−0.08	−0.02	−0.05	−0.05	−0.07	0.08
nRMS	0.68	0.69	0.86	0.78	0.77	0.74	0.90

5.2. MM-CNN-MHA-E Prediction of Geoelectric Fields During the March and December 2015 Geomagnetic Storms

The MM-CNN-MHA-E model was applied to geoelectric field prediction at the Quanzhou observatory during the geomagnetic storms of 17–18 March 2015 and 14–15 December 2015, and the prediction results were compared with those of the persistence model.

First, point-wise evaluation metrics were used to assess the performance of the two models in predicting E_x^{meas} and E_y^{meas} . As summarized in Tables 7 and 8, the overall performances of the MM-CNN-MHA-E model and the persistence model for E_x^{meas} and E_y^{meas} prediction were generally comparable. Although the MM-CNN-MHA-E model achieved slightly better results than the persistence model for several evaluation metrics, the overall differences were not substantial.

As pointed out by Upendran et al. (2022) and Bailey et al. (2022), strong autocorrelation exists in the input data; therefore, point-wise statistical metrics alone may be insufficient to fully distinguish the performance differences between deep learning models and persistence models. Accordingly, event-based evaluation metrics were further employed in this study to assess model performance.

Figure 6 presents the POD, POFD, and HSS metrics for E_x^{meas} obtained from the MM-CNN-MHA-E model and the persistence model. The results indicate that, for E_x^{meas} , the MM-CNN-MHA-E model achieves higher POD at higher thresholds, with overall POD levels exceeding those of the persistence model. At lower thresholds, the MM-CNN-MHA-E model also exhibits lower POFD, thereby outperforming the persistence model. Although the persistence model yields a lower POFD at the 60 mV/km threshold, its POD decreases significantly. The HSS of the MM-CNN-MHA-E model remains consistently higher than that of the persistence model.

Figure 7 shows the POD, POFD, and HSS metrics for E_y^{meas} obtained from the MM-CNN-MHA-E model and the persistence model. For E_y^{meas} , the MM-CNN-MHA-E model similarly demonstrates higher POD at higher thresholds and overall superior performance compared with the persistence model. At lower thresholds, it maintains lower POFD, again showing better overall performance. Notably, at the 50 mV/km threshold, the persistence model produces a POD of 0, indicating an inability to predict events effectively. In contrast, the MM-CNN-MHA-E model achieves a POD of 0.5, demonstrating good predictive capability. Moreover, the HSS of the MM-CNN-MHA-E model also consistently surpasses that of the persistence model. At the 50 mV/km threshold, the MM-CNN-MHA-E model reaches an HSS of 0.12, while the persistence model remains at 0, further illustrating the limited predictive skill of the persistence model under high-threshold conditions.

Overall, the MM-CNN-MHA-E model exhibits significantly stronger geoelectric field prediction capability than the persistence model.

Table 7
RMSE, ME, and nRMS for E_x^{meas} Prediction Using MM-CNN-MHA-E and the Persistence Model

	MM-CNN-MHA-E	Persistence model
RMSE (mV/km)	11.2	11.7
ME (mV/km)	−0.9	−0.3
nRMS	0.21	0.22

Table 8
RMSE, ME, and nRMS for E_y^{meas} Prediction Using MM-CNN-MHA-E and the Persistence Model

	MM-CNN-MHA-E	Persistence model
RMSE (mV/km)	7.6	7.9
ME (mV/km)	0.03	0.05
nRMS	0.28	0.29

5.3. Prediction of GICs During the March and December 2015 Geomagnetic Storms Using MM-CNN-MHA-E and MM-CNN-MHA-GIC

We employed the MM-CNN-MHA-E model to predict the geoelectric field and subsequently computed GICs using physics-based modeling. In addition, the MM-CNN-MHA-GIC model was used to directly predict GICs. Both predicted GIC results were compared with those of the persistence model. For consistency in the analysis, the absolute values of GICs were used throughout, since the sign merely indicates the direction of current flow into or out of the transformer.

As shown in Table 9, when point-wise evaluation metrics were used to compare the GIC prediction capabilities of the MM-CNN-MHA model and the persistence model, the overall performance differences between the two models remained relatively small, with the MM-CNN-MHA model only slightly outperforming the persistence model.

As shown in Figure 8, consistent with the results of Bailey et al. (2022), compared with using MM-CNN-MHA-E to predict the geoelectric field, the accuracy of MM-CNN-MHA-E in predicting GICs is significantly reduced. Figure 8a shows that the PODs of both MM-CNN-MHA-GIC and MM-CNN-MHA-E are higher than those of the persistence model. Figure 8b shows that the HSS of MM-CNN-MHA-E is generally higher than that of

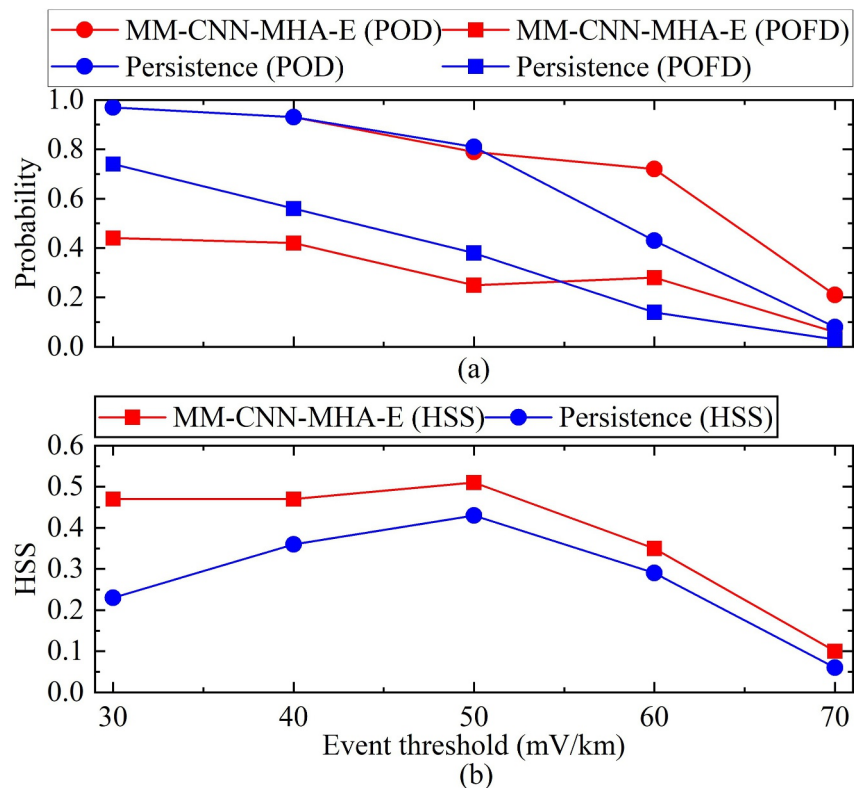


Figure 6. Event-analysis metrics of geoelectric field prediction during the geomagnetic storms of March and December 2015 using the MM-CNN-MHA-E model and the persistence model. (a) POD and POFD for E_x^{meas} , (b) HSS for E_x^{meas} .

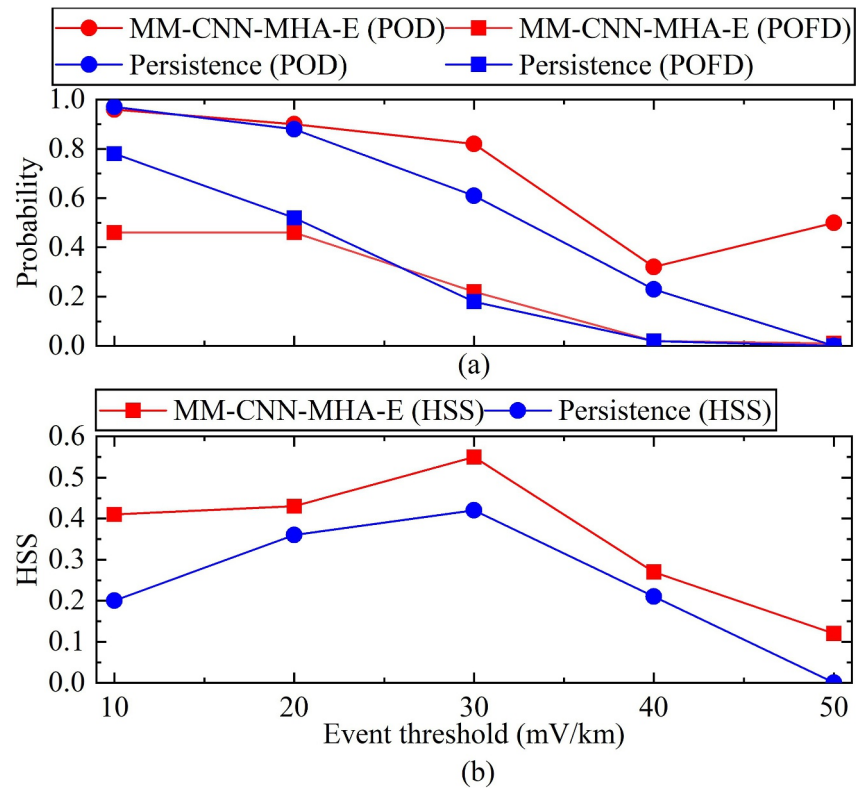


Figure 7. Event-based metrics of geoelectric field prediction during the geomagnetic storms of March and December 2015 using the MM-CNN-MHA-E model and the persistence model. (a) POD and POFD for E_y^{meas} ; (b) HSS for E_y^{meas} .

MM-CNN-MHA-GIC. At nearly all thresholds, both models yield higher HSS values compared to the persistence model.

It should be particularly noted that, although geomagnetic observations were included as input features in this study, the MM-CNN-MHA model still demonstrated better predictive capability than the persistence model. In many previous studies, magnetometer observations are not directly incorporated as input features to avoid predictions becoming overly similar to persistence forecasts. However, the results of this study indicate that, even with geomagnetic observations included as model inputs, the MM-CNN-MHA model generally outperformed the persistence model in terms of POD, POFD, and HSS.

These results suggest that the MM-CNN-MHA model does not merely rely on the continuation of the current geomagnetic state but is also capable of learning the nonlinear coupling relationships among solar wind driving, geomagnetic disturbances, and geoelectric fields, thereby improving overall predictive capability.

5.4. Comparison of Predicted GIC and GIC Calculated Using 1D Conductivity, MT Impedance, and Measured Geoelectric Field During the March and December 2015 Geomagnetic Storms

The GICs predicted by the MM-CNN-MHA-E and MM-CNN-MHA-GIC models were compared with GICs calculated using the 1D conductivity model, MT impedance data, and measured geoelectric fields.

Table 9
RMSE, ME, and nRMS for GIC Prediction Using MM-CNN-MHA and the Persistence Model

	MM-CNN-MHA-E	MM-CNN-MHA-GIC	Persistence model
RMSE (A)	0.26	0.28	0.28
ME (A)	0.02	0.07	0.004
nRMS	0.75	0.83	0.84

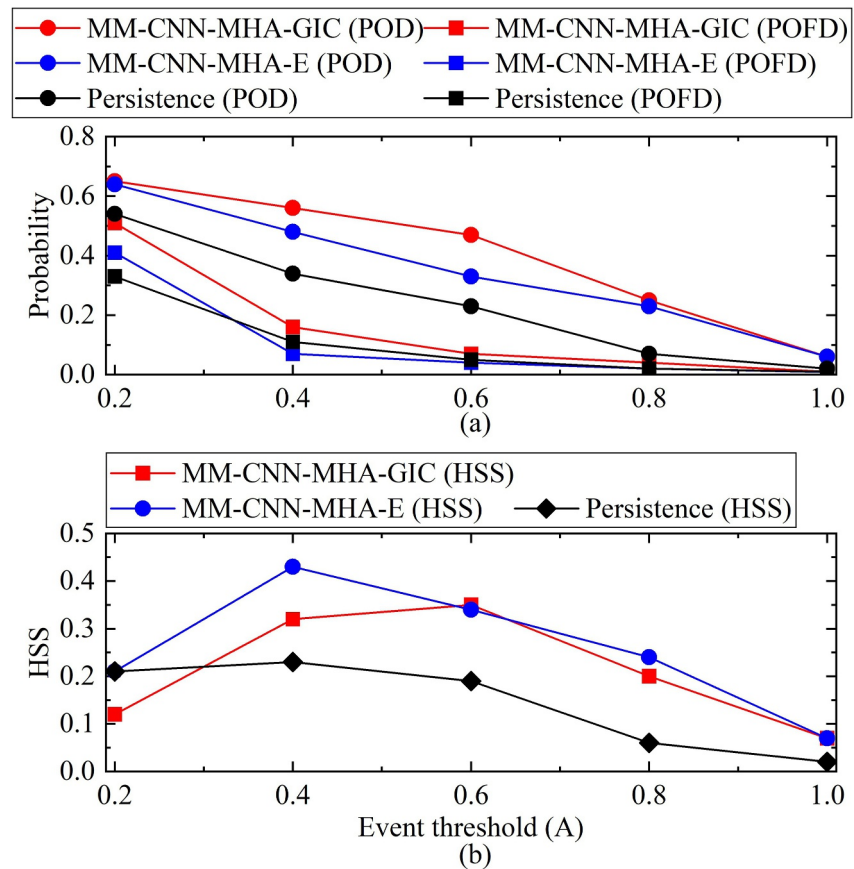


Figure 8. Event-based metrics of GIC prediction during the March and December 2015 geomagnetic storms using MM-CNN-MHA-E, MM-CNN-MHA-GIC, and the persistence model. (a) POD and POFD; (b) HSS.

For the GIC calculations based on the 1D conductivity model and MT impedance data, the measured geomagnetic field B from the Quanzhou geomagnetic observatory was used as the input. The geoelectric fields were subsequently calculated using the 1D conductivity model and MT impedance data, respectively. The induced transmission line voltages were then computed, followed by the calculation of GICs at the HMS substation.

For the MM-CNN-MHA-E model, the geoelectric fields were directly predicted first, after which the transmission line induced voltages and the corresponding GICs at the HMS substation were calculated. Similarly, the GICs derived from measured geoelectric fields were obtained using the observed geoelectric field data from the Quanzhou geoelectric observatory as input, followed by transmission line voltage calculation and subsequent GIC computation at the HMS substation.

As shown in Figure 9, overall, all models can reasonably capture the general trend of the observed GICs. Among them, the GICs predicted by MM-CNN-MHA-E and MM-CNN-MHA-GIC, as well as the GICs calculated from measured geoelectric fields, are closer in magnitude to the observed GICs, whereas the GICs computed from the 1D conductivity model and MT impedance are generally significantly higher than the true values.

Figure 9a shows that both MM-CNN-MHA-E and MM-CNN-MHA-GIC can predict the variation trend of GICs with reasonable accuracy, without exhibiting extreme amplitude deviations. The measured GIC reached a peak of 3.28 A at 04:48 on 17 March 2015 (marked by the black circle), and both models captured the abrupt change at this moment. Specifically, the predicted values were 1.03 A for MM-CNN-MHA-E and 0.6 A for MM-CNN-MHA-GIC (marked by the orange circles). These predictions significantly underestimated the measured peak, mainly because high-amplitude samples were extremely scarce in all data sets: only 7 samples exceeded 1.5 A, 2 samples exceeded 2 A, and only 1 sample exceeded 3 A. Consequently, the models failed to accurately reproduce the peak, a phenomenon consistent with the findings of Bailey et al. (2022). Notably, MM-CNN-MHA-E produced a prediction closer to the measured peak, which is consistent with its better HSS performance under high

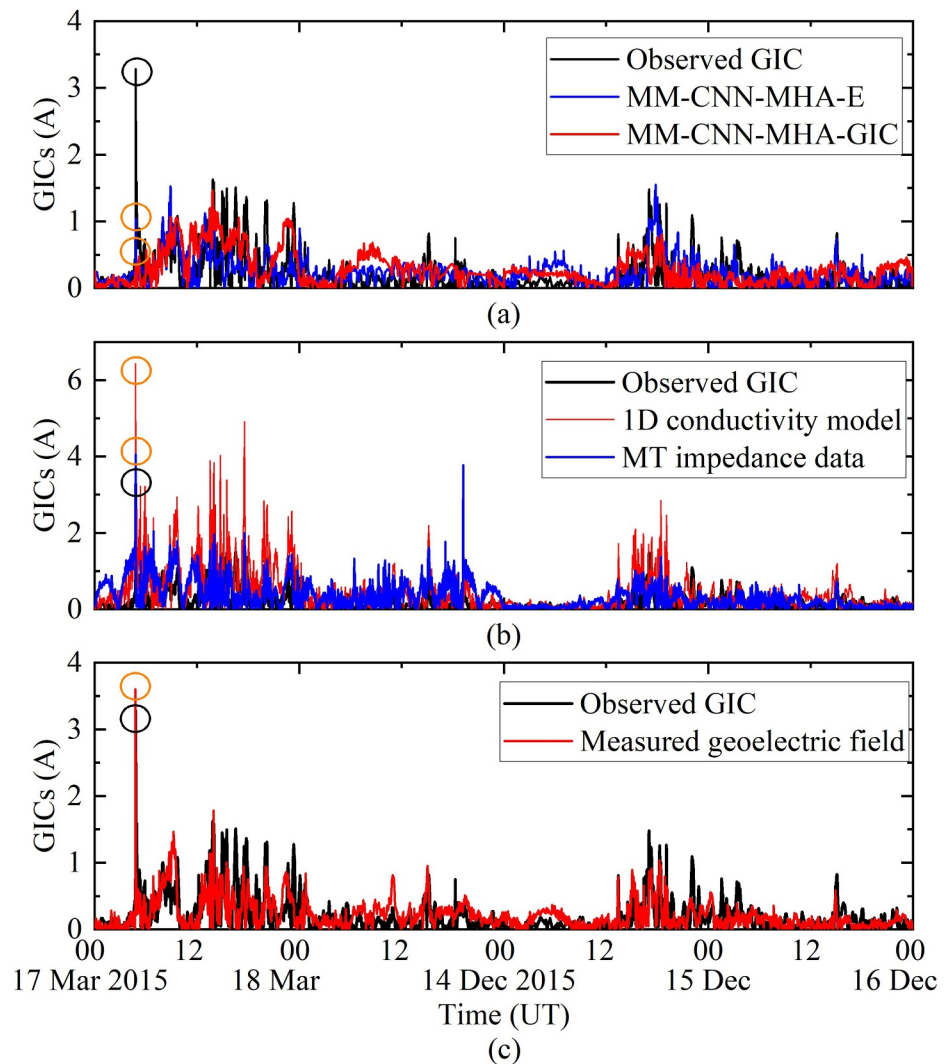


Figure 9. Comparison of GIC time series during the geomagnetic storms of 17–18 March and 14–15 December 2015 (UT). (a) 60-min-ahead GIC predictions obtained from the MM-CNN-MHA-E and MM-CNN-MHA-GIC models; (b) GICs calculated using geomagnetic observations from the Quanzhou observatory combined with the 1D conductivity model and MT impedance data; (c) GICs calculated using the measured geoelectric fields at the Quanzhou observatory.

thresholds, as shown in Section 5.3. Overall, the prediction results of this study are generally consistent with previous research (Bailey et al., 2022; Keese et al., 2020; Upendran et al., 2022), further validating the reliability of our findings.

Figure 9b presents the GICs calculated using a 1D conductivity model and MT impedance measurements. It should be noted that the vertical axis ranges from 0 to 6 A. The results indicate that both models tend to overestimate the GICs. Although both models captured the GIC spike at 04:48 on 17 March 2015, the 1D conductivity model produced a value of 6.43 A, and the MT impedance-based calculation yielded 4.1 A (marked by the orange circles), while the measured value was only 3.28 A (marked by the black circle). Compared with the MT impedance-based results, the GIC amplitudes derived from the 1D conductivity model are significantly higher.

Figure 9c shows that the GICs calculated from the measured geoelectric field are more accurate than the other four models in both amplitude and detail. At 04:48 on 17 March 2015, the GIC calculated from the measured geoelectric field reached 3.6 A (marked by the orange circle), which is very close to the measured value of 3.28 A (marked by the black circle). Overall, no significant overestimation is observed in the calculated results.

Table 10
RMSE, ME, and nRMS of GICs Calculated Using Different Models

	Measured geoelectric fields	MM-CNN-MHA-E	MM-CNN-MHA-GIC	1D conductivity	MT impedance
RMSE (A)	0.24	0.26	0.28	0.52	0.39
ME (A)	0.005	0.02	0.07	0.22	0.14
nRMS	0.7	0.75	0.83	1.54	1.18

Table 10 summarizes the RMSE, ME, and nRMS metrics of the GICs calculated using different methods. The GICs derived from measured geoelectric fields exhibited the best overall performance, with the lowest RMSE (0.24 A), the smallest ME (0.005 A), and the lowest nRMS (0.70). These results indicate that this approach achieved not only the highest overall prediction accuracy but also the smallest systematic bias.

The MM-CNN-MHA-E model performed better than the MM-CNN-MHA-GIC model in terms of RMSE and nRMS. Regarding systematic bias, the ME value of the MM-CNN-MHA-E model was 0.02 A, smaller than the 0.07 A obtained by the MM-CNN-MHA-GIC model. This suggests that the geoelectric field-driven prediction approach exhibits lower systematic bias, with its ME approaching that of the GICs directly calculated from measured geoelectric fields.

In contrast, the GICs calculated using MT impedance data exhibited larger RMSE (0.39 A), nRMS (1.18), and ME (0.14 A) values, indicating reduced overall prediction accuracy and increased systematic bias. Among all methods, the 1D conductivity model showed the poorest performance, with RMSE, nRMS, and ME values reaching 0.52 A, 1.54, and 0.22 A, respectively, indicating a pronounced overestimation tendency.

As discussed in Section 4.2, because the empirical coefficients a and b were calibrated using the 17–18 March 2015 geomagnetic storm, and this event was subsequently used for model performance evaluation, a certain degree of potential data leakage may exist for the March 2015 event. To assess whether this issue affects the model performance evaluation, we further calculated the GIC prediction results for different models using only the independent 14–15 December 2015 geomagnetic storm, which was not used for empirical-coefficient calibration. As shown in Table S5 of Supporting Information S1, the model performance ranking obtained from the December 2015 event is consistent with that obtained from the combined March and December 2015 events. The GICs derived from measured geoelectric fields exhibited the best overall performance, with the lowest RMSE (0.17 A), the smallest ME (−0.003 A), and the lowest nRMS (0.68). The MM-CNN-MHA-E model performed better than the MM-CNN-MHA-GIC model in terms of RMSE and nRMS. In contrast, the GICs calculated using MT impedance data exhibited larger RMSE (0.23 A), nRMS (0.91), and ME (0.07 A), indicating reduced overall prediction accuracy and increased systematic bias. Among all methods, the 1D conductivity model showed the poorest performance, with RMSE, nRMS, and ME values of 0.32 A, 1.26, and 0.11 A, respectively, indicating a pronounced overestimation tendency. These results suggest that the potential data leakage associated with the March 2015 event has a limited impact on the model performance evaluation and does not alter the main scientific conclusions of this study.

In addition, the GIC prediction results summarized in Table 10 were compared with the geomagnetic disturbance and GIC prediction results obtained using the Space Weather Modeling Framework (SWMF) model in Zhang et al. (2020). The two studies are comparable because they employed the same study region, including the Quanzhou geomagnetic observatory and the HMS 220-kV substation in Fujian Province, China, as well as the same observed GIC data during the geomagnetic storms.

It should be noted that Zhang et al. (2020) mainly analyzed the nRMS values between the SWMF-predicted geomagnetic disturbances and observations, with the predicted nRMS values being approximately 0.5 for B_x and 1.1 for B_y . In the present study, the SWMF geomagnetic disturbance data provided by Zhang et al. (2020) were further used together with the same GIC calculation method and parameters reported in their study to recalculate the nRMS between the predicted and observed GICs. The results indicate that the nRMS of the SWMF-based predicted GICs was approximately 0.85.

By comparison, the nRMS values of the five GIC prediction methods in this study ranged from 0.70 to 1.54. Among them, the GICs calculated using measured geoelectric fields and those predicted by the MM-CNN-MHA models achieved nRMS values ranging from 0.70 to 0.83, which are comparable to and better than the SWMF-

based results. In contrast, the GICs calculated using the 1D conductivity model and MT impedance data yielded nRMS values ranging from 1.18 to 1.54, indicating overall poorer performance than the SWMF results.

As shown in Table S6 of Supporting Information S1, the performances of different models were not fully consistent when evaluated using event-based metrics. Overall, the GICs calculated from measured geoelectric fields and the MM-CNN-MHA-based models generally maintained relatively high HSS values under most threshold conditions and achieved a favorable balance between POD and POFD, indicating relatively stable overall event detection capability.

The MT impedance model exhibits lower POFD values under higher-threshold conditions while maintaining moderate HSS values, suggesting a conservative predictive tendency and fewer false alarms for strong GIC events. However, its POD values decrease as thresholds increase, indicating limited capability to capture strong events. In addition, the relatively large RMSE, ME, and nRMS values indicate biases in point-wise GIC prediction. Although the 1D conductivity model achieves higher POD and HSS values under most thresholds, its higher POFD and larger point-wise errors indicate a stronger tendency toward overprediction and false alarms.

6. Discussion

6.1. Comparison Between GIC Predictions From MM-CNN-MHA and Physics-Based GIC Calculations

As shown in Section 5.4, the GICs predicted by the MM-CNN-MHA models were generally closer to the GICs calculated from measured geoelectric fields than those obtained using the 1D conductivity model and MT impedance methods. This is mainly because the MM-CNN-MHA-E model directly predicts measured geoelectric fields, whereas the MM-CNN-MHA-GIC model predicts GICs fitted from measured geoelectric field observations. Consequently, both deep learning approaches show relatively good agreement with the GICs calculated from measured geoelectric fields.

The Quanzhou geoelectric observatory is located approximately 280 km from the HMS substation, and the two locations may correspond to different subsurface conductivity structures. Despite this spatial inconsistency, the proposed models still demonstrated good GIC prediction capability, indicating that the models are able, to some extent, to learn and capture the dominant features and temporal evolution characteristics governing GIC variations. This further suggests that the proposed approach has potential for GIC prediction at target substations under practical conditions where geoelectric field observations are relatively sparse.

In addition, the GIC calculation errors based on the 1D conductivity model and MT impedance model were generally larger than those of the deep learning models. This is mainly because conventional physics-based models strongly depend on the accuracy of the subsurface conductivity structure. The 1D conductivity assumption is often oversimplified under complex geological conditions and may not adequately represent realistic subsurface structures. Furthermore, factors such as the limited spatial distribution of MT observation sites and observational noise may also introduce uncertainties into impedance estimation. In contrast, deep learning approaches are capable of learning complex nonlinear relationships from multisource data, thereby reducing, to some extent, the impacts of inaccurate conductivity models and insufficient MT observations on GIC prediction results.

Taken together, the results demonstrate that the GICs calculated using locally measured geoelectric fields achieved the highest accuracy and outperformed those calculated using locally measured geomagnetic fields combined with Earth impedance models. This result is physically expected because the geoelectric field is the direct driver of GICs. It further highlights the critical importance of geoelectric field observations for GIC monitoring, model validation, and operational forecasting. Compared with estimating geoelectric fields from geomagnetic observations and Earth conductivity models, direct geoelectric field measurements avoid additional uncertainties associated with assumptions regarding subsurface conductivity structure, impedance estimation errors, and the spatial variability of geomagnetic fields, thereby improving the accuracy of GIC estimation and prediction.

6.2. Comparison Between GIC Predictions From MM-CNN-MHA-E and MM-CNN-MHA-GIC

As shown in Section 5.4, the MM-CNN-MHA-E model, which predicts geoelectric fields first and subsequently calculates GICs, achieved higher prediction accuracy than the MM-CNN-MHA-GIC model, which directly predicts GICs.

From the perspective of the underlying physical mechanisms, GICs are not directly driven by geomagnetic disturbances but by geoelectric fields. Therefore, geoelectric fields serve as the key physical variable linking space weather processes and power-grid responses. Direct GIC prediction is essentially an end-to-end modeling approach, whereas the strategy of first predicting geoelectric fields and subsequently calculating GICs decomposes the forecasting process into two stages: “driving source prediction” and “system response calculation.” Consequently, this framework is more consistent with the physical processes governing GIC generation. Preserving the physical transfer chain from solar wind disturbances to geomagnetic variations, geoelectric fields, and ultimately GICs, therefore, helps maintain the physical consistency of the prediction process.

In addition to its physical consistency, the two approaches also differ in terms of error propagation. Direct GIC prediction must simultaneously learn the complex mapping involving uncertainties associated with geoelectric fields, power-grid parameters, and observational noise, making the learning task more challenging. In contrast, the MM-CNN-MHA-E framework concentrates the primary prediction uncertainty in the geoelectric field prediction stage and subsequently calculates GICs using a relatively stable physics-based power-grid model. This decomposition reduces error accumulation and improves the overall prediction accuracy.

Furthermore, the two models are trained using different target variables. The MM-CNN-MHA-E model is trained directly using measured geoelectric field observations, whereas the MM-CNN-MHA-GIC model is trained using proxy GICs derived from empirical fitting relationships. Consequently, the MM-CNN-MHA-E model benefits from higher-fidelity training data while preserving the physical calculation process from geoelectric fields to GICs. In contrast, the MM-CNN-MHA-GIC model bypasses this physical process. This further contributes to the superior performance of the MM-CNN-MHA-E model and is likely one of the major reasons why it outperforms the MM-CNN-MHA-GIC model.

Taken together, these results demonstrate that, compared with direct end-to-end GIC prediction, preserving the physical transfer chain from solar wind disturbances to geomagnetic variations, geoelectric fields, and ultimately GICs leads to more stable and reliable prediction results. This finding indicates that geoelectric fields are the key intermediate physical variable in space weather-driven GIC prediction, and improving the accuracy of geoelectric field prediction is an important pathway toward enhancing GIC prediction performance. More importantly, the results suggest that, compared with purely data-driven end-to-end GIC prediction, a hybrid forecasting framework that combines machine learning with physics-based power-grid models is better able to maintain the physical consistency of the prediction process and provides an important reference for future operational forecasting of space weather-driven GICs.

6.3. Operational Relevance

Although different GIC prediction models were systematically compared in this study, some conclusions remain subject to limitations in practical engineering applications because the overall GIC amplitudes in the Fujian 220-kV power grid were relatively low (approximately 0–3 A). As discussed in Section 5.4, the GICs calculated using the 1D conductivity model exhibited a noticeable overestimation tendency, with a predicted peak value of approximately 6.43 A, whereas the corresponding observed GIC peak was only 3.28 A. However, even the predicted peak value of 6.43 A remains substantially lower than the reference levels commonly considered in the North American Electric Reliability Corporation (NERC) TPL-007 transmission reliability standard (approximately 75–85 A/phase). Therefore, under the conditions investigated in this study, the differences in prediction errors among the models are still insufficient to directly correspond to practical operational warning or protection-action issues.

In addition, under low-GIC conditions, the error characteristics exhibited by different prediction models may not necessarily represent their performance under high-GIC conditions. Moreover, slight temporal shifts in model responses may produce relatively large errors in point-wise metrics such as RMSE, even though the models may still accurately capture GIC peaks and their temporal evolution within short time windows. Therefore, conclusions regarding GIC prediction accuracy should still be interpreted with caution.

Overall, the 1D conductivity model and MT impedance method generally tended to overestimate GIC amplitudes, whereas the MM-CNN-MHA models exhibited a certain underestimation tendency. These findings may provide useful references for future GIC studies in regions characterized by higher GIC amplitudes.

6.4. Challenges and Improvements in GIC Prediction

This study performed GIC prediction for the HMS substation in Fujian Province, China. The results indicate that GIC prediction remains highly challenging. For example, the models were unable to accurately reproduce the GIC variations at every time step, and the overall predicted amplitudes were generally lower than the observations. These issues are also among the major challenges reported in current GIC prediction studies (Bailey et al., 2022; Keese et al., 2020; Upendran et al., 2022).

The above discrepancies may mainly originate from several factors. First, machine learning models lack sufficient continuous observed GIC data for training, making it difficult for the models to fully learn the true variation characteristics of GICs. Meanwhile, GICs derived from empirical fitting or physics-based modeling are simultaneously affected by multiple uncertainties, including geomagnetic fields, subsurface impedance structures, geoelectric fields, and power grid parameters. Second, solar wind parameters, geomagnetic observations, and geoelectric field observations all contain observational noise, which may further affect the reliability of model training and prediction results. In addition, the minute-level temporal resolution adopted in this study may lead to the loss of part of the high-frequency disturbance information, thereby contributing to the underestimation of GIC amplitudes.

Compared with previous studies, several improvements were introduced in the prediction framework of this study. First, geomagnetic disturbance observations were incorporated into the input features to more comprehensively represent the coupling processes among the solar wind, magnetosphere, ionosphere, and ground environment. Second, measured geoelectric fields were adopted as the prediction target, which is more physically consistent with the driving mechanism of GICs. In addition, a multi-module feature extraction strategy was introduced to reduce the impacts of data noise on prediction performance.

Through these improvements, the GIC prediction results obtained in this study were overall superior to those calculated using the 1D conductivity model and MT impedance model. This indicates that incorporating geomagnetic disturbance observations and geoelectric field observations is important for improving GIC prediction capability. Future studies should further integrate observed GIC data and magnetosphere–ionosphere-related observations, such as satellite-based magnetospheric observations, to continuously improve the reliability and accuracy of GIC prediction.

7. Summary

In this study, the HMS 220 kV substation in Fujian Province was selected as the study site to investigate space weather–driven GIC forecasting. Two GIC prediction frameworks based on the MM-CNN-MHA model were developed. The first framework (MM-CNN-MHA-E) predicts the geoelectric field at the Quanzhou geoelectric observatory and subsequently calculates GICs at the HMS substation using a physics-based power-grid model. The second framework (MM-CNN-MHA-GIC) directly predicts GICs using proxy GICs derived from measured geoelectric field observations as the training targets. Both frameworks were validated using the March and December 2015 geomagnetic storms, and their prediction performance was systematically compared with GICs calculated from measured geoelectric fields, the 1D conductivity model, and the MT impedance model.

The results demonstrate that the proposed MM-CNN-MHA framework provides effective forecasting of space weather–driven GICs. However, because the Quanzhou geomagnetic/geoelectric observatory is located approximately 280 km from the HMS substation, the measured geoelectric field at Quanzhou cannot fully represent the actual geoelectric field at the HMS substation owing to the influence of the complex subsurface resistivity structure in Fujian Province. This finding further suggests that geomagnetic and geoelectric field observations should, where possible, be deployed closer to high-risk substations to obtain more representative local observations. Overall, the main conclusions of this study are as follows:

1. Compared with several commonly used deep learning models, the MM-CNN-MHA model improves the representation of the nonlinear relationships among solar wind parameters, geomagnetic disturbances, and geoelectric fields by grouping the input features according to their physical properties and extracting features using multiple modules. These results demonstrate the potential of the proposed model for GIC forecasting.
2. The MM-CNN-MHA-E framework, which first predicts geoelectric fields and subsequently calculates GICs, consistently outperforms the MM-CNN-MHA-GIC framework, which directly predicts GICs. This result indicates that preserving the physical chain from solar wind input to geomagnetic disturbances, geoelectric

- fields, and ultimately GICs leads to more stable and reliable GIC forecasting. Compared with purely data-driven end-to-end GIC prediction, the hybrid framework combining deep learning with a physics-based power-grid model better preserves physical consistency and improves GIC prediction accuracy.
- GICs calculated using measured geoelectric fields outperform those calculated using measured geomagnetic fields combined with either the 1D conductivity model or the MT impedance model. These results demonstrate the critical role of local geoelectric field observations in GIC monitoring, model validation, and operational forecasting. Direct acquisition of high-quality geoelectric field observations can effectively reduce GIC estimation errors associated with uncertainties in subsurface resistivity structure.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The solar wind observation data were obtained from the OMNIWeb database provided by the NASA Goddard Space Flight Center (GSFC) (https://omniweb.gsfc.nasa.gov/ow_min.html). The geomagnetic field data used in this study were obtained from the China Earthquake Networks Center, National Earthquake Data Center (https://data.earthquake.cn/D13000/index_1.html). The geoelectric field data used in this study were obtained from the China Earthquake Networks Center, National Earthquake Data Center (<https://data.earthquake.cn/datashare/report.shtml?PAGEID=datasourcelist&dt=40280d0453e581090153e583c50b0001>). Registration with the National Earthquake Data Center is required to access the geomagnetic and geoelectric field data. The corresponding data pages provide contact information for the data provider, which can be used for inquiries regarding data access. The measured magnetotelluric impedance data for Fujian Province were obtained from Xu et al. (2019). The 1D conductivity model of Fujian Province and the GIC observation data at the Huangmeishan substation were also provided by Zhang et al. (2020). The GIC calculation program was provided by Cordell et al. (2021). The GIC data and solar wind parameters supporting the findings of this study are available in Zhang (2025a). The data processing programs and the GIC prediction software used in this manuscript (version 1.0.0) are available in Zhang (2025b).

Acknowledgments

We would like to thank the China Earthquake Networks Center, National Earthquake Data Center for providing the geomagnetic disturbance and geoelectric field data from the Quanzhou station. The authors also appreciate the State Grid Corporation of China for supplying the 220 kV power grid parameters in Ningde, Fujian Province. Special thanks are extended to Prof. Jiao Jiao Zhang from the National Space Science Center, Chinese Academy of Sciences, for providing the 1D conductivity model of Fujian Province and the GIC observation data from the Huangmeishan substation. This work is jointly supported by the China Magnetotelluric Array (Phase I), National Science and Technology Major Project (2024ZD1000202), the National Natural Science Foundation of China (U24A20602, 42504063) and the Natural Science Foundation of Chongqing (CSTB2024NSCQ-JQX0040, CSTB2024NSCQ-LZX0011). We also sincerely thank the Associate Editor and the two anonymous reviewers for their constructive comments and valuable suggestions, which greatly improved the quality and clarity of this manuscript.

References

- Bailey, R. L., Leonhardt, R., Möstl, C., Beggan, C., Reiss, M. A., Bhaskar, A., & Weiss, A. J. (2022). Forecasting GICs and geoelectric fields from solar wind data using LSTMs: Application in Austria. *Space Weather*, 20(3), e2021SW002907. <https://doi.org/10.1029/2021SW002907>
- Beggan, C. D., Richardson, G. S., Baillie, O., Hübert, J., & Thomson, A. W. P. (2021). Geoelectric field measurement, modelling and validation during geomagnetic storms in the UK. *Journal of Space Weather and Space Climate*, 11, 37. <https://doi.org/10.1051/swsc/2021022>
- Blais, G., & Metsa, P. (1993). Operating the Hydro-Quebec grid under magnetic storm conditions since the storm of 13 March 1989. In *Paper presented at Solar-Terrestrial Predictions*.
- Bolduc, L. (2002). GIC observations and studies in the Hydro-Québec power system. *Journal of Atmospheric and Solar-Terrestrial Physics*, 64(16), 1793–1802. [https://doi.org/10.1016/S1364-6826\(02\)00128-1](https://doi.org/10.1016/S1364-6826(02)00128-1)
- Boteler, D. H., & Pirjola, R. J. (2017). Modeling geomagnetically induced currents. *Space Weather*, 15(1), 258–276. <https://doi.org/10.1002/2016SW001499>
- Chen, H., Ren, Z., & Tang, J. (2025). Magnetotelluric adaptive inversion using multi-resolution tetrahedral grids: Application to the North China craton. *Journal of Geophysical Research: Solid Earth*, 130(4), e2024JB030405. <https://doi.org/10.1029/2024JB030405>
- Collado-Villaverde, A., Muñoz, P., & Cid, C. (2021). Deep neural networks with convolutional and LSTM layers for SYM-H and ASY-H forecasting. *Space Weather*, 19(6), e2021SW002748. <https://doi.org/10.1029/2021SW002748>
- Cordell, D., Mann, I. R., Parry, H., Unsworth, M. J., Cui, R. Y., Clark, C., et al. (2024). Modeling geomagnetically induced currents in the Alberta power network: Comparison and validation using Hall probe measurements during a magnetic storm. *Space Weather-the International Journal of Research and Applications*, 22(4), e2023SW003813. <https://doi.org/10.1029/2023sw003813>
- Cordell, D., Unsworth, M. J., Lee, B., Haneson, C., Milling, D. K., & Mann, I. R. (2021). Estimating the geoelectric field and electric power transmission line voltage during a geomagnetic storm in Alberta, Canada using measured magnetotelluric impedance data: The influence of three-dimensional electrical structures in the lithosphere. *Space Weather*, 19(10), e2021SW002803. <https://doi.org/10.1029/2021SW002803>
- Gaunt, C. T., & Coetzee, G. (2007). Transformer failures in regions incorrectly considered to have low GIC-risk. In *Paper presented at 2007 IEEE Lausanne Power Tech*.
- Honkoniemi, I., Kuvshinov, A., Rastätter, L., & Pulkkinen, A. (2018). Predicting global ground geoelectric field with coupled geospace and three-dimensional geomagnetic induction models. *Space Weather*, 16(8), 1028–1041. <https://doi.org/10.1029/2018SW001859>
- Hübert, J., Beggan, C. D., Richardson, G. S., Gomez-Perez, N., Collins, A., & Thomson, A. W. P. (2024). Validating a UK geomagnetically induced current model using differential magnetometer measurements. *Space Weather*, 22(2), e2023SW003769. <https://doi.org/10.1029/2023SW003769>
- Keesee, A. M., Pinto, V., Coughlan, M., Lennox, C., Mahmud, M. S., & Connor, H. K. (2020). Comparison of deep learning techniques to model connections between solar wind and ground magnetic perturbations. *Frontiers in Astronomy and Space Sciences*, 7–2020, 550874. <https://doi.org/10.3389/fspas.2020.550874>

- Kelbert, A., & Lucas, G. M. (2020). Modified GIC estimation using 3-D earth conductivity. *Space Weather*, 18(8), e2020SW002467. <https://doi.org/10.1029/2020SW002467>
- Lehtinen, M., & Pirjola, R. (1985). Currents produced in earthed conductor networks by geomagnetically-induced electric-fields. *Annales Geophysicae*, 3(4), 479–484. <http://pascal-francis.inist.fr/vibad/index.php?action=getRecordDetail&idt=8580990>
- Liu, C., Li, Y., & Pirjola, R. (2014). Observations and modeling of GIC in the Chinese large-scale high-voltage power networks. *Journal of Space Weather and Space Climate*, 4, A03. <https://doi.org/10.1051/swsc/2013057>
- Liu, T. Z., Shi, X., Hartinger, M. D., Angelopoulos, V., Rodger, C. J., Viljanen, A., et al. (2024). Global observations of geomagnetically induced currents caused by an extremely intense density pulse during a coronal mass ejection. *Space Weather*, 22(10), e2024SW003993. <https://doi.org/10.1029/2024SW003993>
- Long, D., Chen, Y., Toth, G., Zou, S., Pulkkinen, T., Ren, J., et al. (2022). New findings from explainable SYM-H forecasting using gradient boosting machines. *Space Weather*, 20(8), e2021SW002928. <https://doi.org/10.1029/2021SW002928>
- Lotz, S. I., & Cilliers, P. J. (2015). A solar wind-based model of geomagnetic field fluctuations at a mid-latitude station. *Advances in Space Research*, 55(1), 220–230. <https://doi.org/10.1016/j.asr.2014.09.014>
- Lotz, S. I., Heyns, M. J., & Cilliers, P. J. (2017). Regression-based forecast model of induced geoelectric field. *Space Weather*, 15(1), 180–191. <https://doi.org/10.1002/2016SW001518>
- Pirjola, R. (2002). Fundamentals about the flow of geomagnetically induced currents in a power system applicable to estimating space weather risks and designing remedies. *Journal of Atmospheric and Solar-Terrestrial Physics*, 64(18), 1967–1972. [https://doi.org/10.1016/S1364-6826\(02\)00228-6](https://doi.org/10.1016/S1364-6826(02)00228-6)
- Pirjola, R. (2005). Effects of space weather on high-latitude ground systems. *Advances in Space Research*, 36(12), 2231–2240. <https://doi.org/10.1016/j.asr.2003.04.074>
- Pratscher, K. M., Ingham, M., Mac Manus, D. H., Kruglyakov, M., Heise, W., Rodger, C. J., et al. (2024). Modeling GIC in the southern South Island of Aotearoa New Zealand using magnetotelluric data. *Space Weather*, 22(7), e2024SW003907. <https://doi.org/10.1029/2024SW003907>
- Priyadarshi, S. (2024). Prediction of high latitude nightside geomagnetic field disturbances (GGFDs) during the 9-March-2012 geomagnetic storm using recurrent neural network (RNN). *Advances in Space Research*, 74(4), 1868–1882. <https://doi.org/10.1016/j.asr.2024.05.036>
- Pulkkinen, A., Lindahl, S., Viljanen, A., & Pirjola, R. (2005). Geomagnetic storm of 29–31 October 2003: Geomagnetically induced currents and their relation to problems in the Swedish high-voltage power transmission system. *Space Weather*, 3(8). <https://doi.org/10.1029/2004SW001013>
- Pulkkinen, A., Pirjola, R., & Viljanen, A. (2007). Determination of ground conductivity and system parameters for optimal modeling of geomagnetically induced current flow in technological systems. *Earth Planets and Space*, 59(9), 999–1006. <https://doi.org/10.1186/BF03352040>
- Pulkkinen, A., Rastätter, L., Kuznetsova, M., Singer, H., Balch, C., Weimer, D., et al. (2013). Community-wide validation of geospace model ground magnetic field perturbation predictions to support model transition to operations. *Space Weather*, 11(6), 369–385. <https://doi.org/10.1002/swe.20056>
- Temerin, M., & Li, X. (2006). Dst model for 1995–2002. *Journal of Geophysical Research*, 111(A4). <https://doi.org/10.1029/2005JA011257>
- Torta, J. M., Serrano, L., Regué, J. R., Sánchez, A. M., & Roldán, E. (2012). Geomagnetically induced currents in a power grid of northeastern Spain. *Space Weather*, 10(6). <https://doi.org/10.1029/2012SW000793>
- Tóth, G., Meng, X., Gombosi, T. I., & Rastätter, L. (2014). Predicting the time derivative of local magnetic perturbations. *Journal of Geophysical Research: Space Physics*, 119(1), 310–321. <https://doi.org/10.1002/2013JA019456>
- Upendran, V., Tigas, P., Ferdousi, B., Bloch, T., Cheung, M. C. M., Ganju, S., et al. (2022). Global geomagnetic perturbation forecasting using deep learning. *Space Weather*, 20(6), e2022SW003045. <https://doi.org/10.1029/2022SW003045>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., et al. (2017). Attention is all you need. In *Proceedings of the 31st international conference on neural information processing systems* (pp. 6000–6010). Curran Associates Inc.
- Wintoft, P., Wik, M., & Viljanen, A. (2015). Solar wind driven empirical forecast models of the time derivative of the ground magnetic field. *Journal of Space Weather and Space Climate*, 5, A7. <https://doi.org/10.1051/swsc/2015008>
- Xu, S., Unsworth, M. J., Hu, X., & Mooney, W. D. (2019). Magnetotelluric evidence for asymmetric simple shear extension and lithospheric thinning in South China. *Journal of Geophysical Research: Solid Earth*, 124(1), 104–124. <https://doi.org/10.1029/2018JB016505>
- Zainuddin, A., Hairuddin, M. A., Iffah, Z., Abd Latiff, Z. I., Anuar, N., Benavides, I., & Yassin, A. I. M. (2024). Prediction of geomagnetically induced currents in low-latitude regions using deep learning. *Current Science*, 127(6), 691. <https://doi.org/10.18520/cs/v127/i6/691-700>
- Zhang, J. J., Wang, C., Sun, T. R., Liu, C. M., & Wang, K. R. (2015). GIC due to storm sudden commencement in low-latitude high-voltage power network in China: Observation and simulation. *Space Weather*, 13(10), 643–655. <https://doi.org/10.1002/2015SW001263>
- Zhang, J. J., Yu, Y. Q., Wang, C., Du, D., Wei, D., & Liu, L. G. (2020). Measurements and simulations of the geomagnetically induced currents in low-latitude power networks during geomagnetic storms. *Space Weather*, 18(8), e2020SW002549. <https://doi.org/10.1029/2020SW002549>
- Zhang, X. J. (2025a). GIC data and solar wind parameters from March and December 2015 in Fujian, China (Version 1) [Dataset]. *Zenodo*. <https://doi.org/10.5281/zenodo.20380712>
- Zhang, X. J. (2025b). Xuejianzhang/predicting GIC package: Version 1.0.0 [Software]. *Zenodo*. <https://doi.org/10.5281/zenodo.20381320>